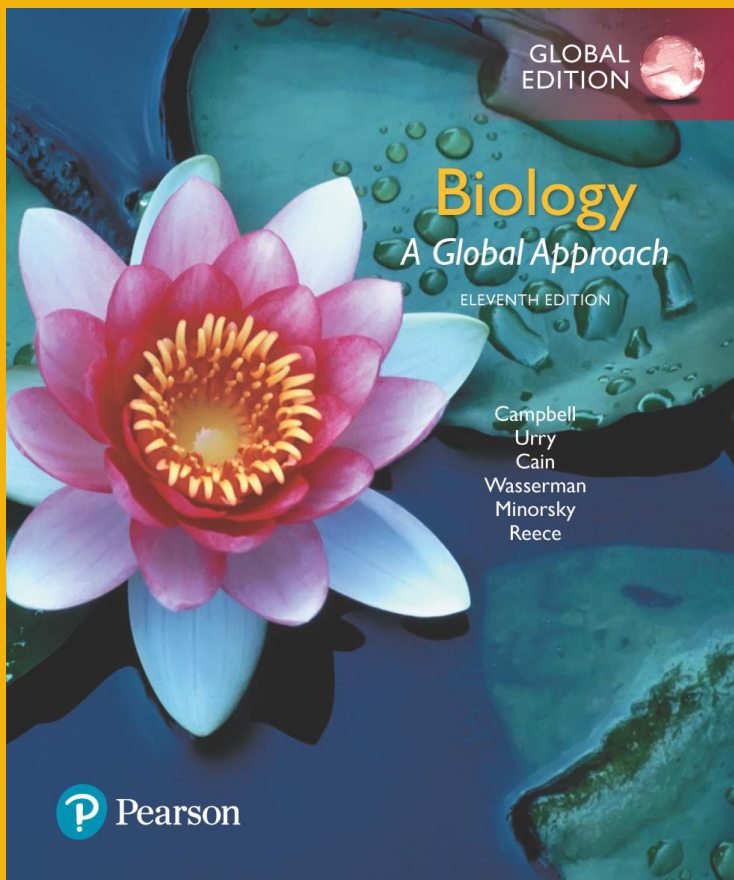




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Chapter 19

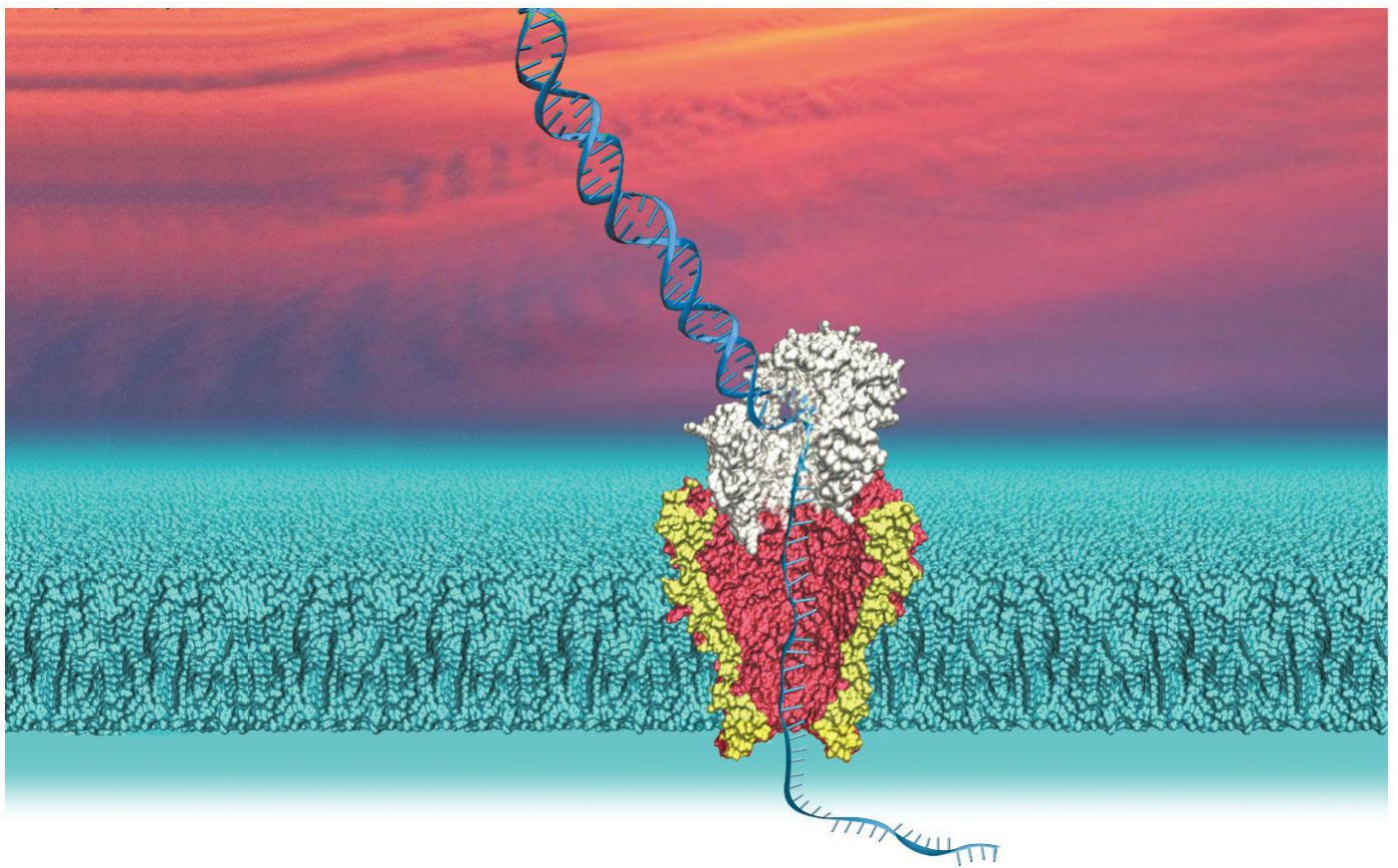
DNA Technology

Lecture Presentations by
Nicole Tunbridge and
Kathleen Fitzpatrick

The DNA Toolbox

- Recently, the genome sequences of two extinct species—Neanderthals and woolly mammoths—have been completed
- Advances in sequencing techniques make genome sequencing increasingly faster and less expensive

Figure 19.1



- The main technologies for sequencing and manipulating DNA are called **DNA technology**
- **Biotechnology** refers to the manipulation of organisms or their components to make useful products

Concept 19.1: DNA sequencing and DNA cloning are valuable tools for genetic engineering and biological inquiry

- The complementarity of the two DNA strands is the basis for **nucleic acid hybridization**, the base pairing of one strand of nucleic acid to the complementary sequence on another strand
- **Genetic engineering** is the direct manipulation of genes for practical purposes

DNA Sequencing

- A gene's complete nucleotide sequence can be determined using a process called **DNA sequencing**
- The first automated procedure was based on a technique called dideoxy or chain termination sequencing, developed by Frederick Sanger
- In the last 15 years, “next-generation sequencing” techniques have been developed that are much faster

Figure 19.2



- In sequencing by synthesis, many DNA fragments are copied
- A specific strand of each fragment is immobilized and the complementary strand synthesized one nucleotide at a time
- Thousands or hundreds of thousands of fragments about 300 nucleotides long can be sequenced in parallel
- This is an example of “high-throughput” technology

Figure 19.3

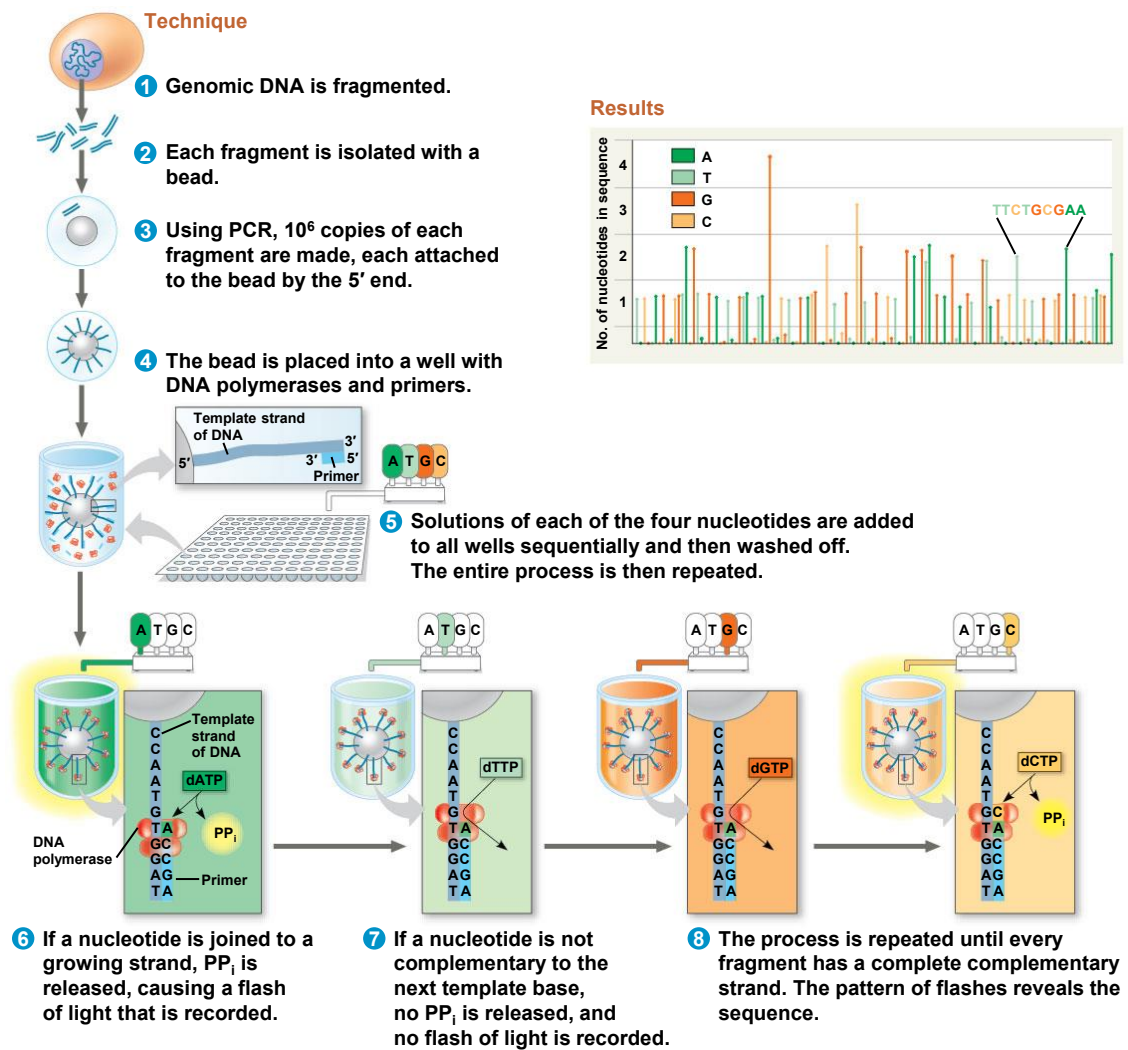


Figure 19.3a

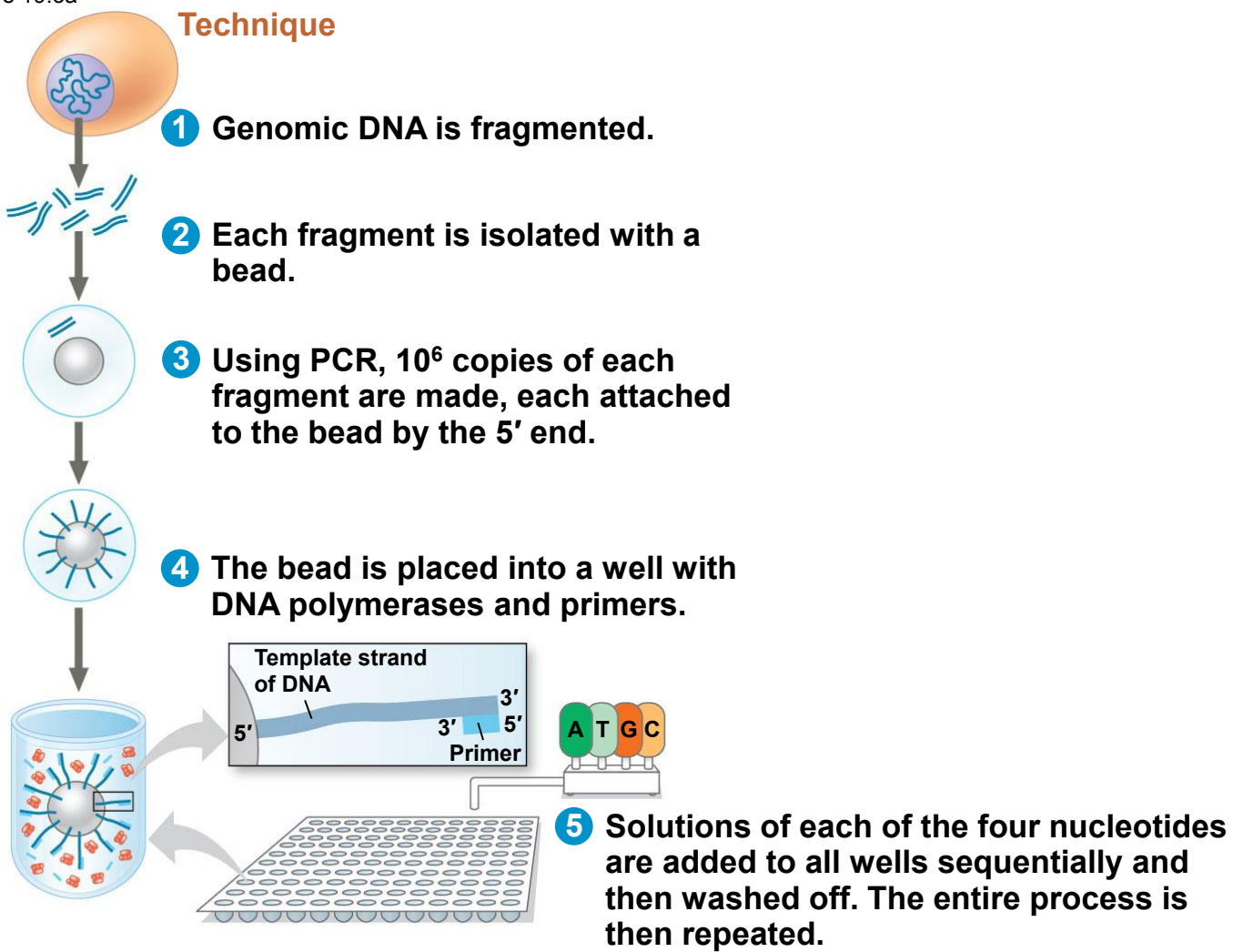


Figure 19.3b

Technique

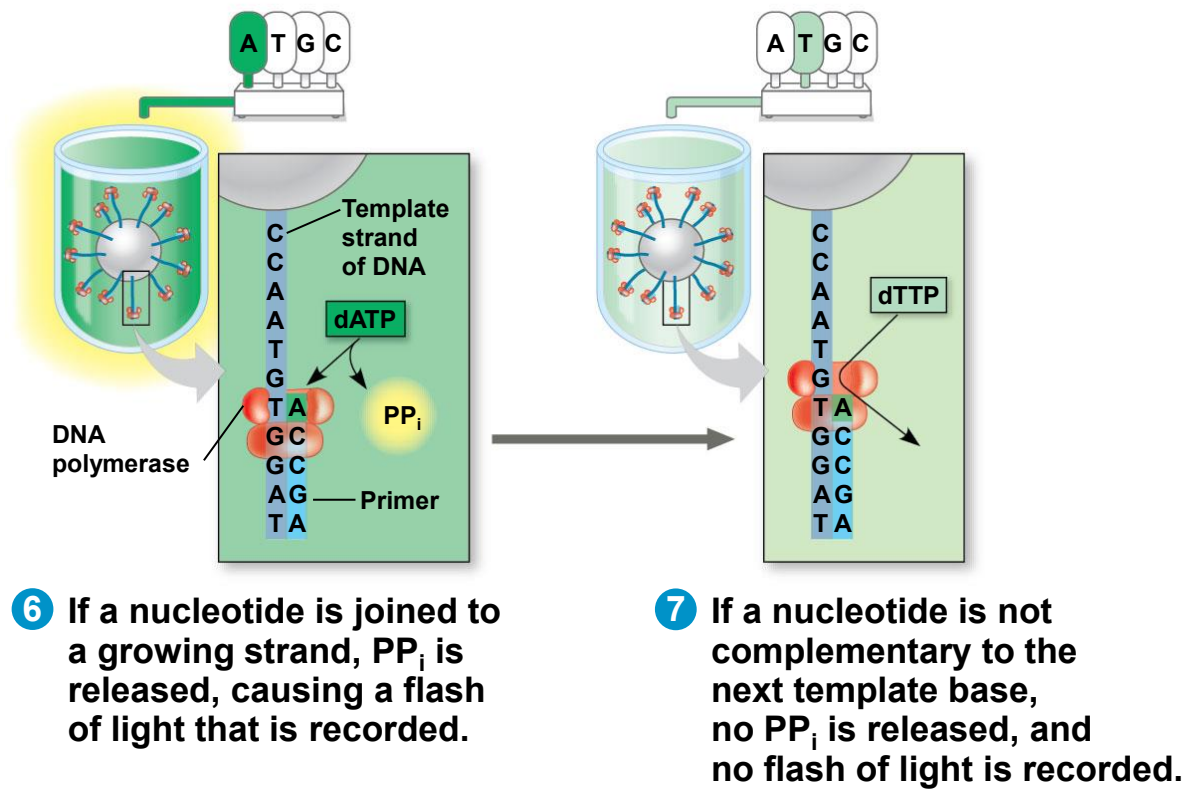
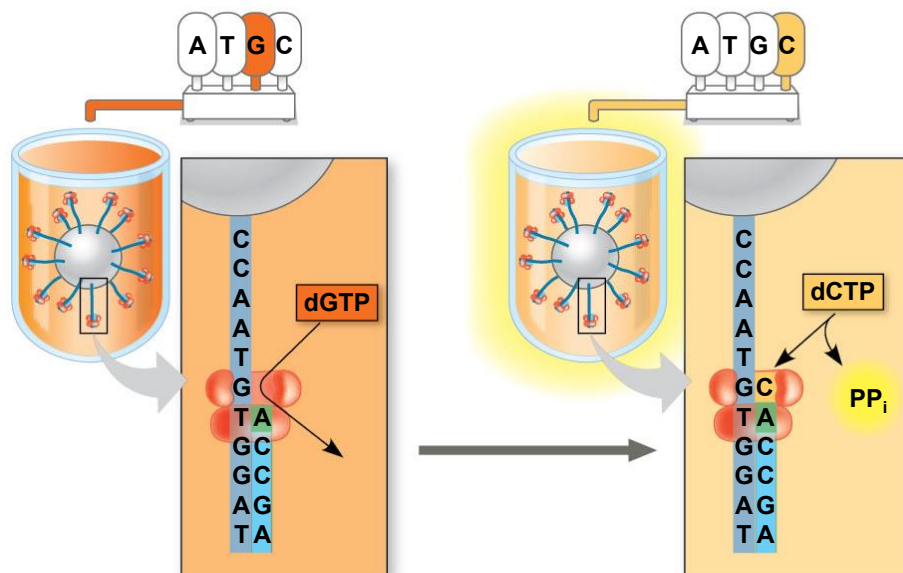


Figure 19.3c

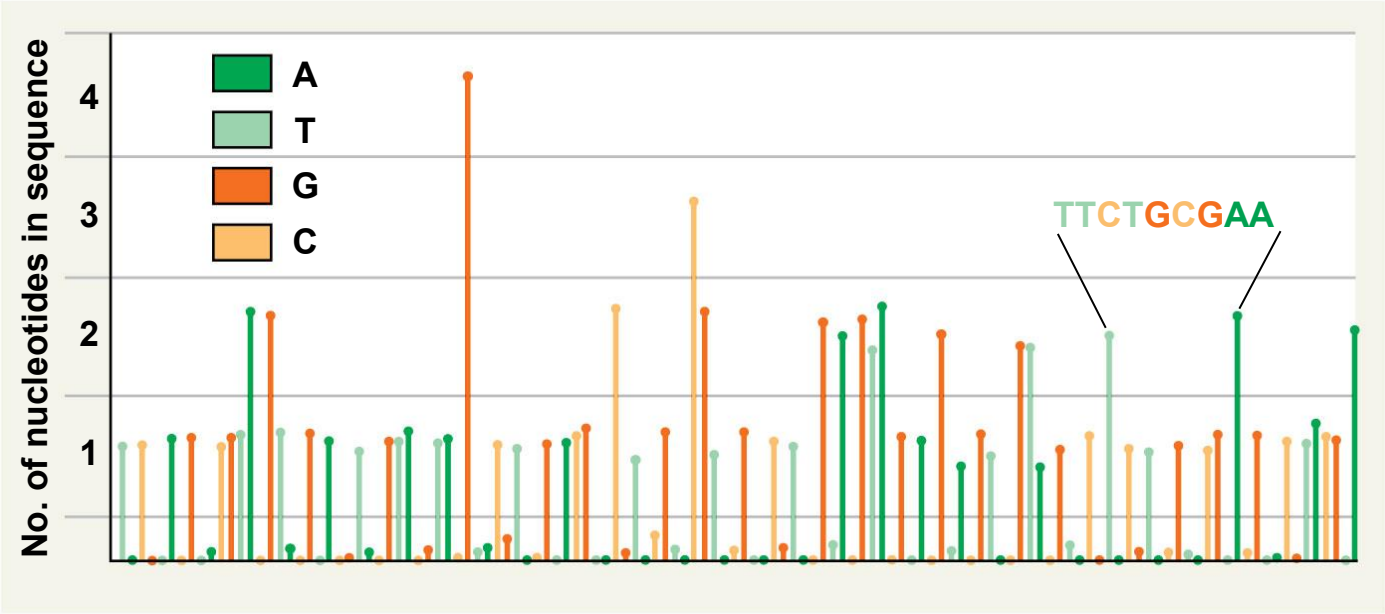
Technique



- 8** The process is repeated until every fragment has a complete complementary strand. The pattern of flashes reveals the sequence.

Figure 19.3d

Results



- In “third-generation sequencing,” the techniques used are even faster and less expensive
- In some methods, a single long DNA molecule is sequenced as it moves through a very small pore (nanopore)
- In one such approach, each base is identified by the way it interrupts an electric current as it passes through the pore

Making Multiple Copies of a Gene or Other DNA Segment

- To work directly with specific genes, scientists prepare well-defined DNA segments in multiple identical copies by a process called **DNA cloning**
- **Plasmids** are small, circular DNA molecules that replicate separately from the bacterial chromosome
- Researchers can insert DNA into a plasmid to produce a **recombinant DNA molecule**, which contains DNA from two different sources

- Reproduction of a recombinant plasmid in a bacterial cell results in cloning of the plasmid including the foreign DNA
- This production of multiple copies of a single gene is a type of DNA cloning called **gene cloning**

Figure 19.4

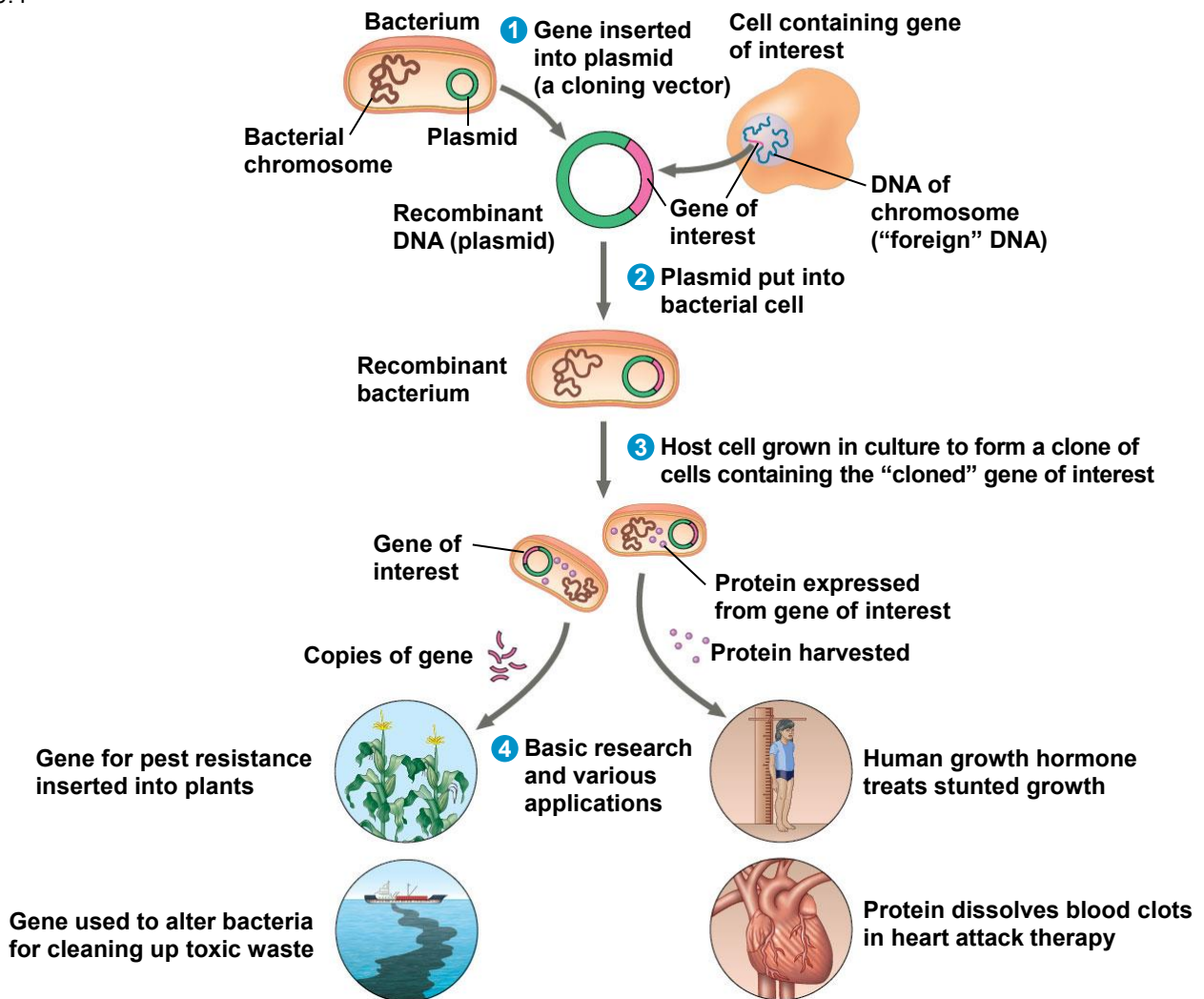


Figure 19.4a

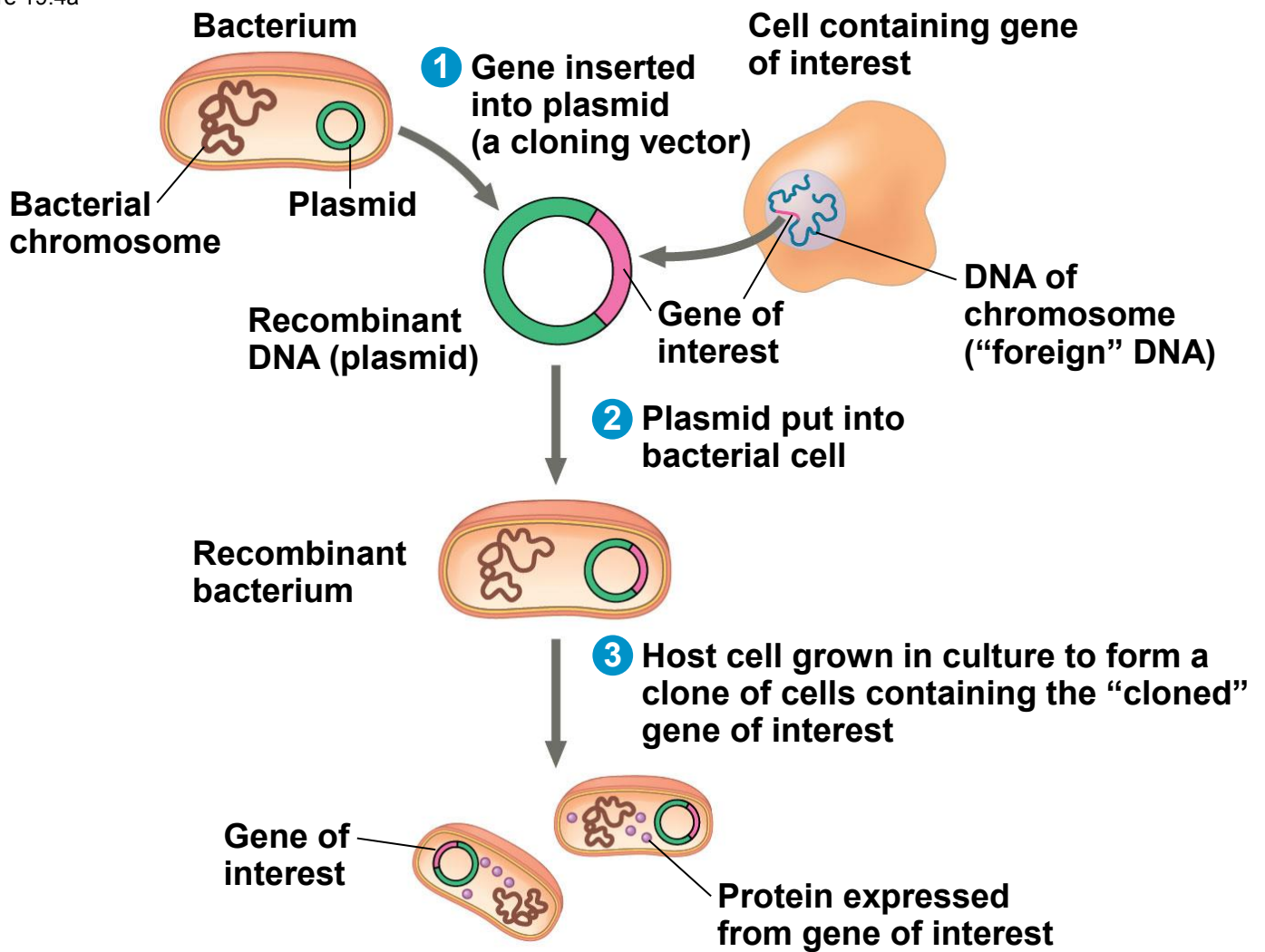
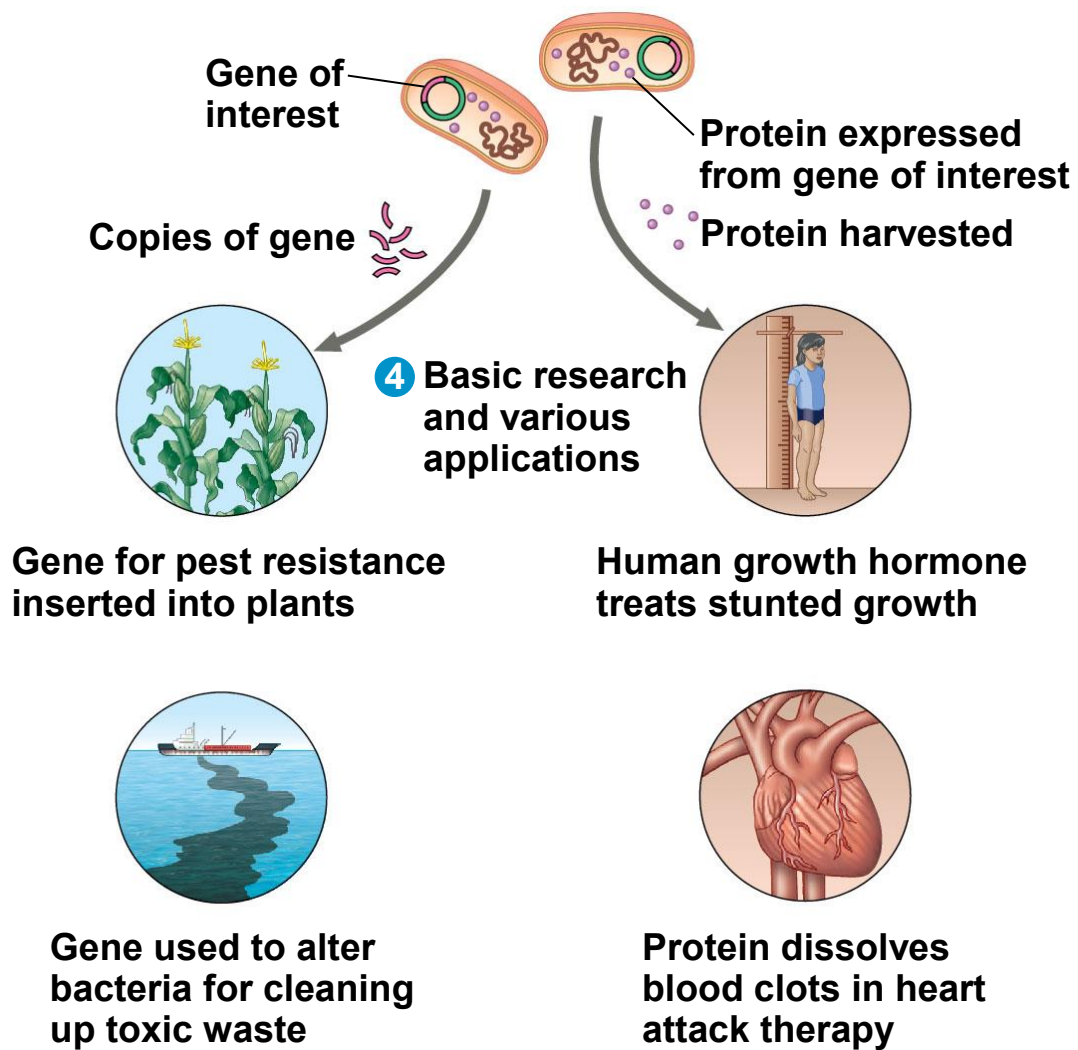


Figure 19.4b



- A plasmid used to clone a foreign gene is called a **cloning vector**
- Bacterial plasmids are widely used as cloning vectors because they are readily obtained, easily manipulated, and easily introduced into bacterial cells, and once in the bacteria they multiply rapidly
- Gene cloning is useful for amplifying genes to produce a protein product for research, medical, or other purposes

Using Restriction Enzymes to Make a Recombinant DNA Plasmid

- Bacterial **restriction enzymes** cut DNA molecules at specific DNA sequences called **restriction sites**
- A restriction enzyme usually makes many cuts, yielding **restriction fragments**
- The most useful restriction enzymes cut DNA in a staggered way, producing fragments with at least one single-stranded end called a **sticky end**

- Sticky ends can bond with complementary sticky ends of other fragments
- DNA ligase is an enzyme that seals the bonds between restriction fragments
- This allows researchers to join two DNA fragments from different sources

Animation: Restriction Enzymes

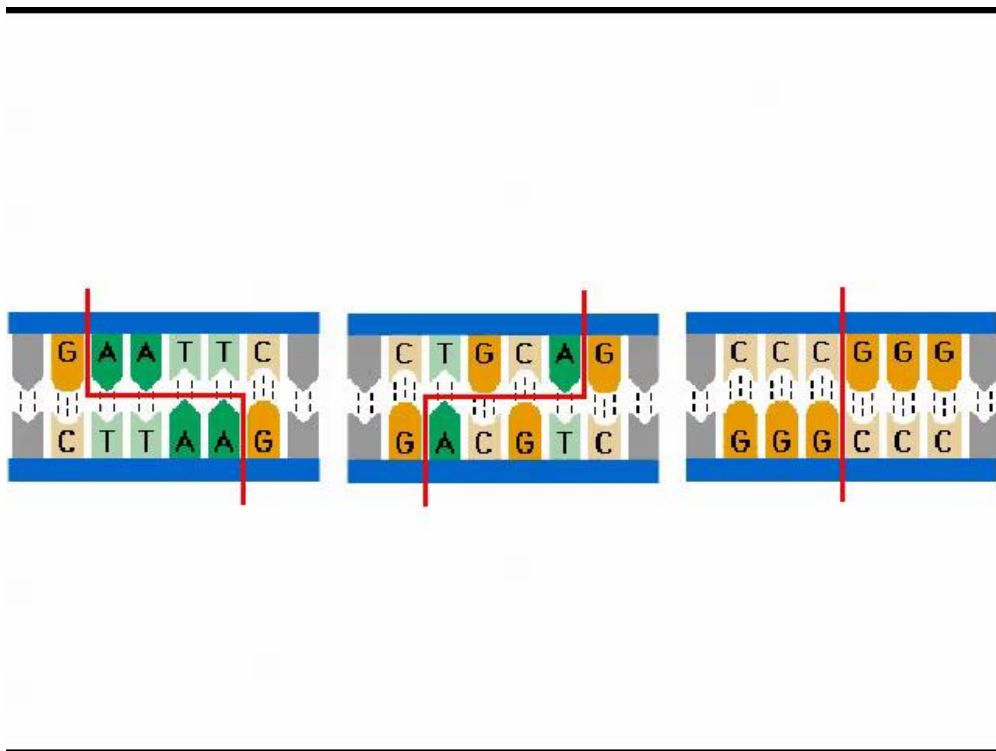


Figure 19.5

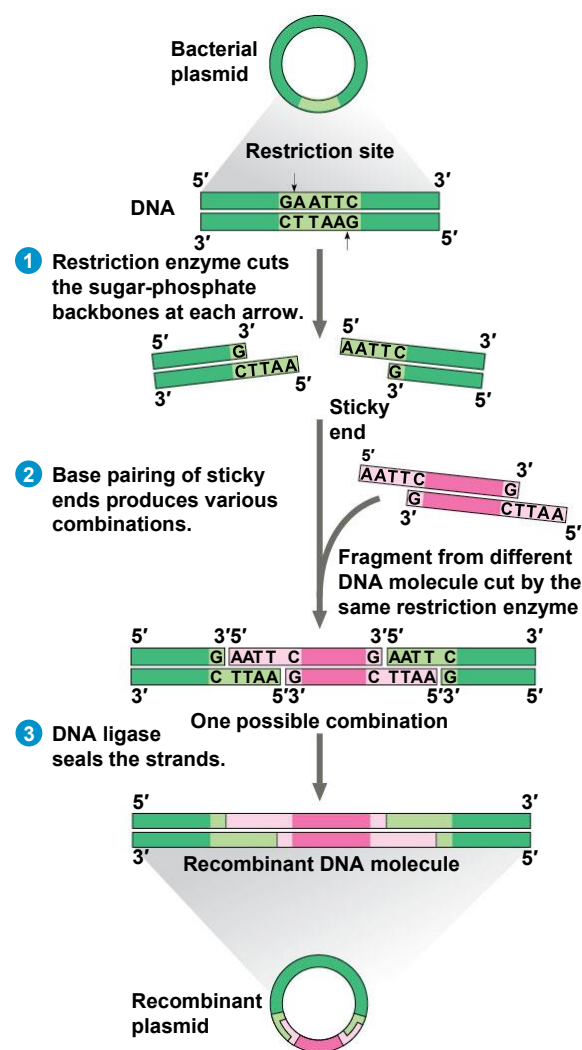


Figure 19.5a

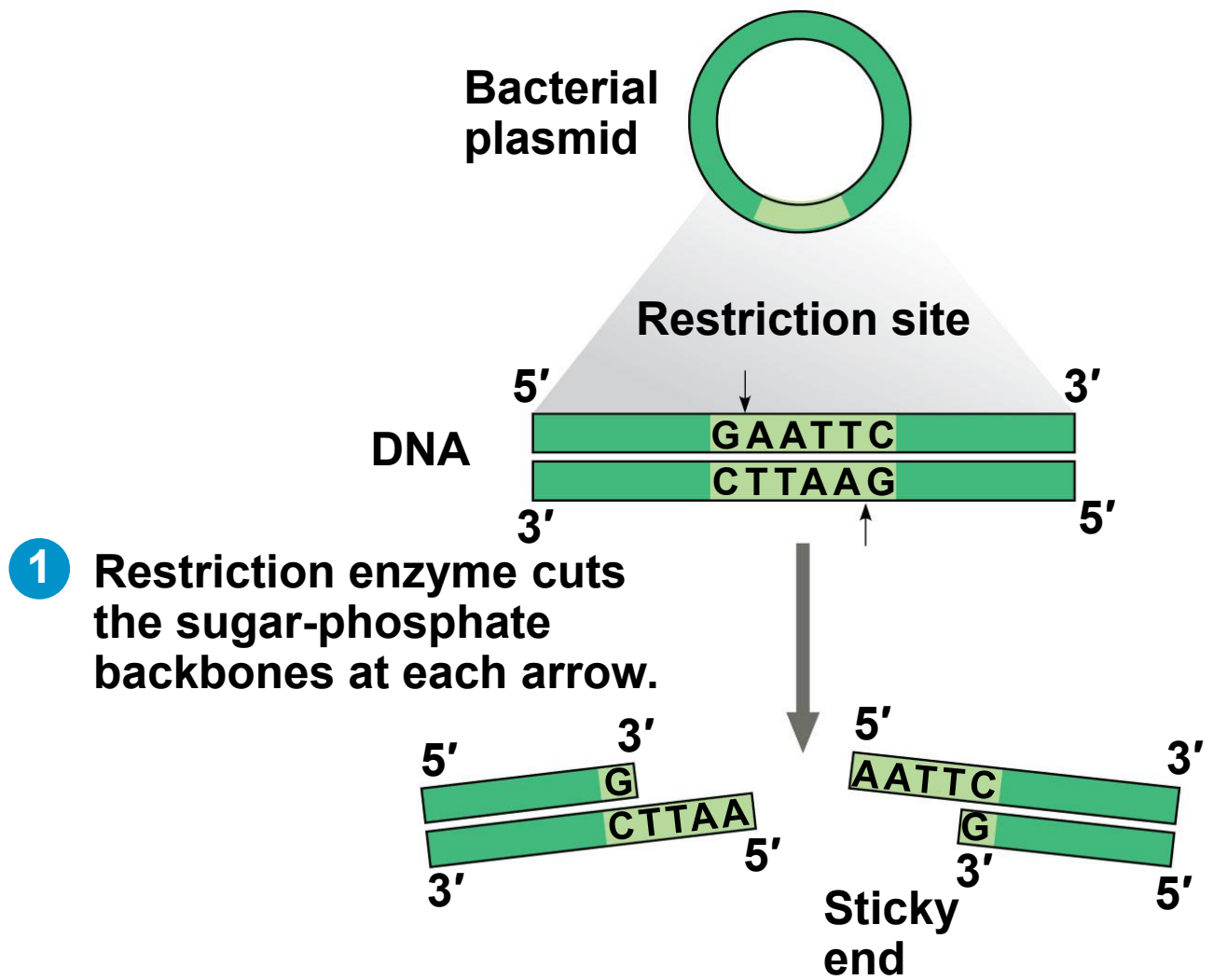


Figure 19.5b

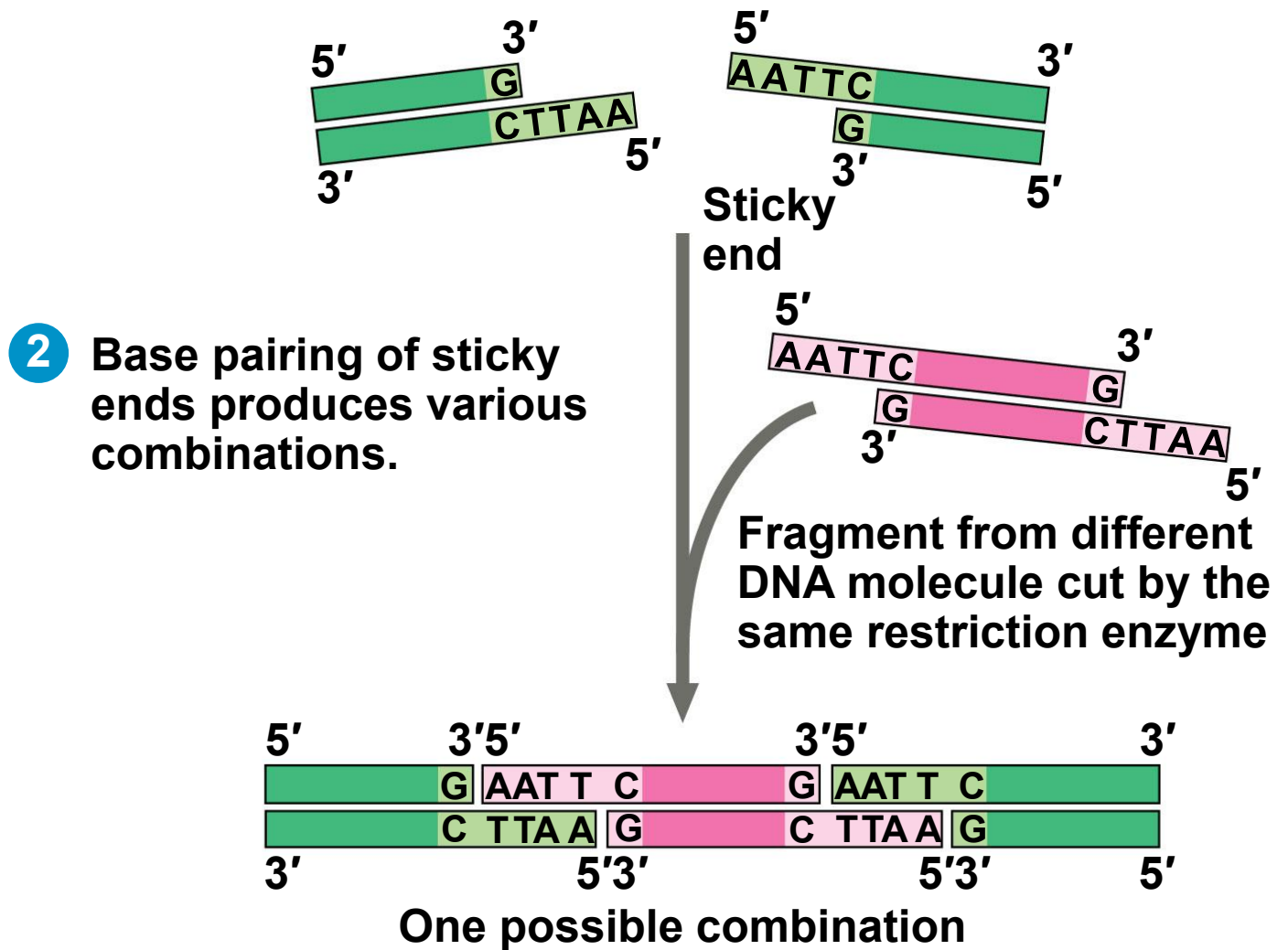
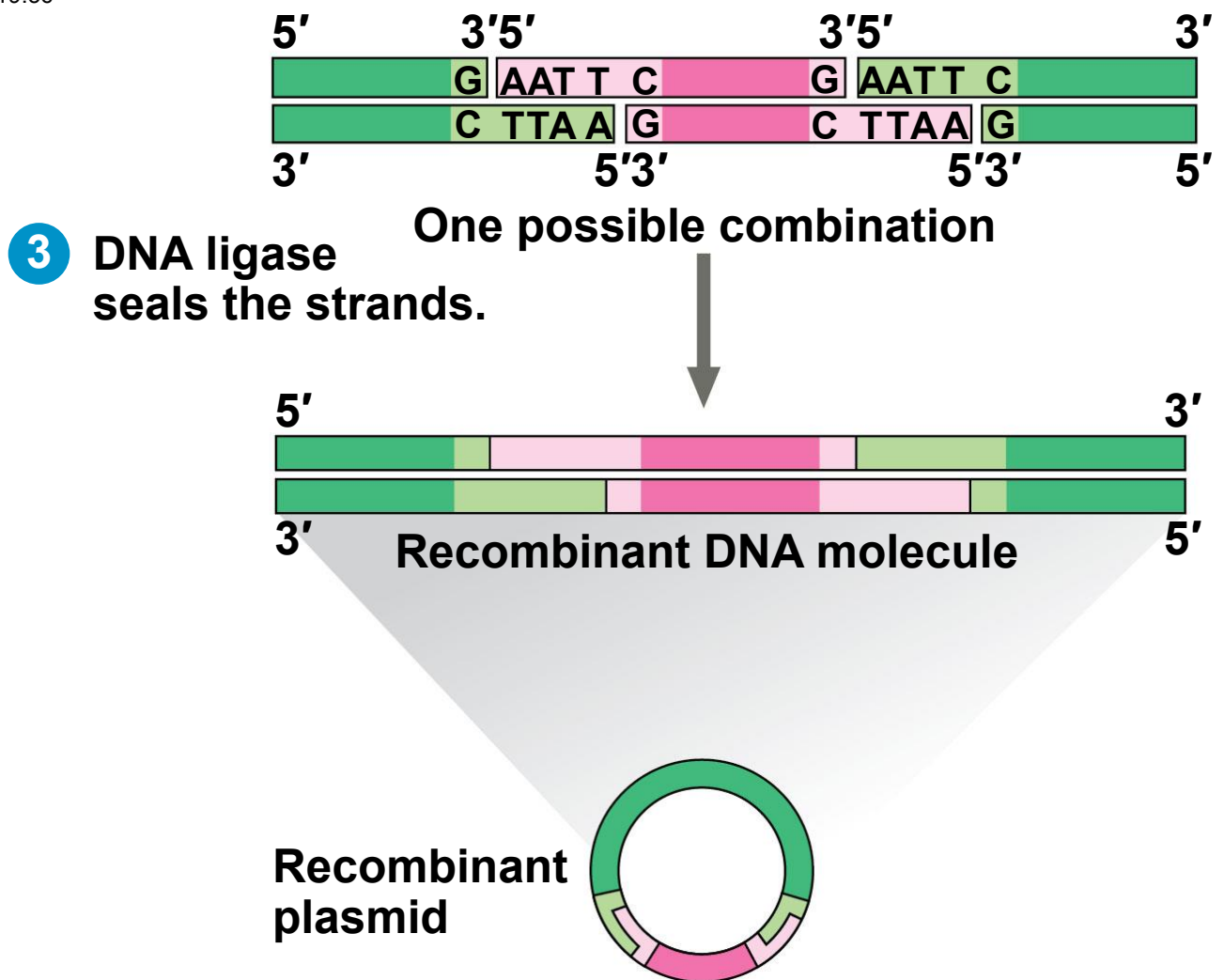
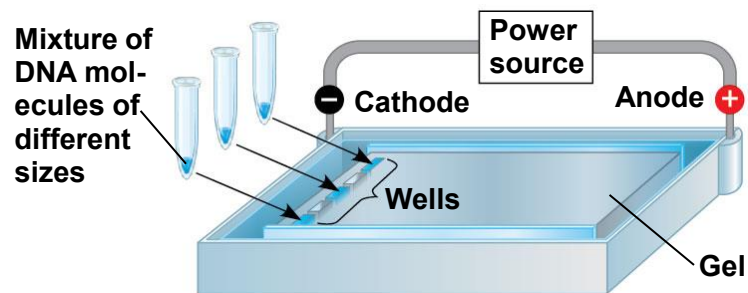


Figure 19.5c

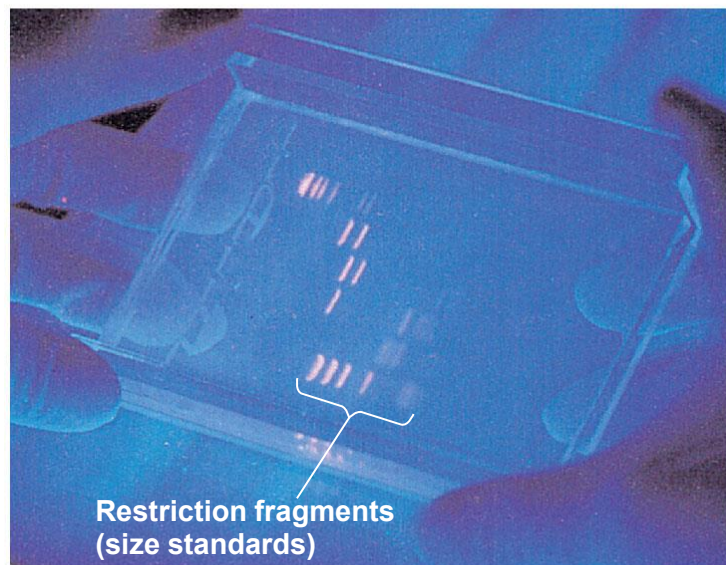


- To check the recombinant plasmid, researchers might cut the products again using the same restriction enzyme
- To separate and visualize the fragments produced, **gel electrophoresis** is carried out
- This technique uses a gel made of a polymer to separate a mixture of nucleic acids or proteins based on size, charge, or other physical properties

Figure 19.6



(a) Negatively charged DNA molecules will move toward the positive electrode.



(b) Shorter molecules are slowed down less than longer ones, so they move faster through the gel.

Amplifying DNA: The Polymerase Chain Reaction (PCR) and Its Use in DNA Cloning

- The **polymerase chain reaction, PCR**, can produce many copies of a specific target segment of DNA
- A three-step cycle—heating, cooling, and replication—brings about a chain reaction that produces an exponentially growing population of identical DNA molecules

- The key to PCR is an unusual, heat-stable DNA polymerase called *Taq* polymerase
- Other polymerases may be used as well; some are more accurate and stable than *Taq*, such as *Pfu* polymerase
- PCR uses a pair of primers specific for the sequence to be amplified
- PCR amplification occasionally incorporates errors into the amplified strands and so cannot substitute for gene cloning in cells

Figure 19.7

Technique

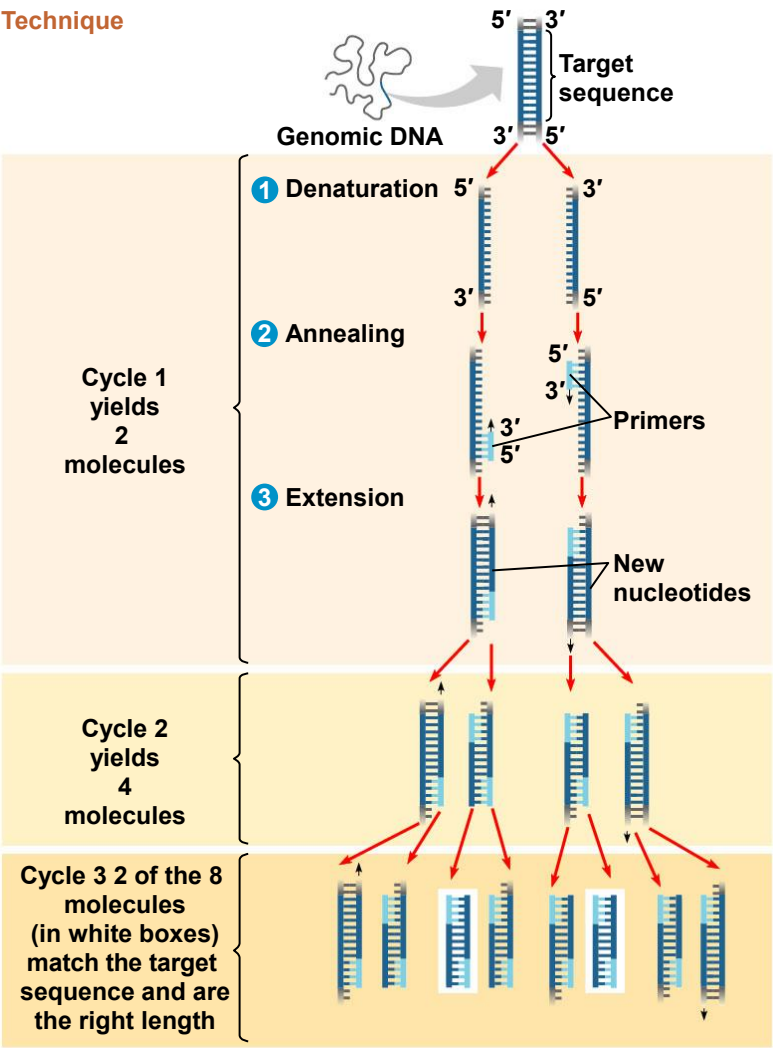


Figure 19.7a

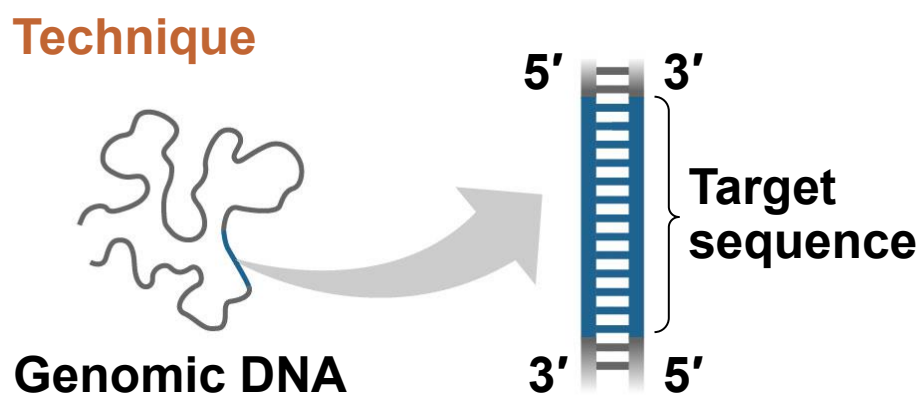


Figure 19.7b_1

Technique

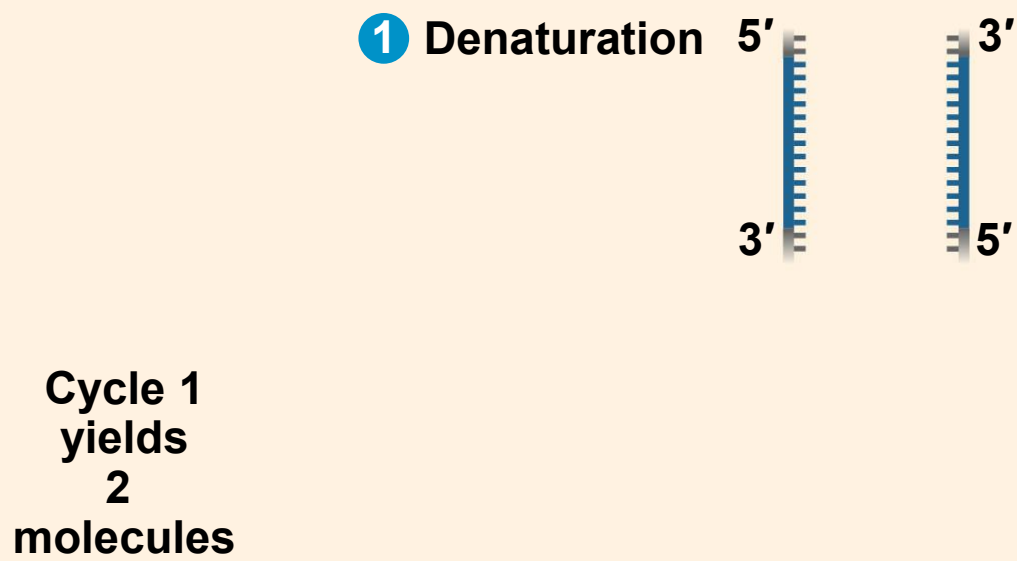


Figure 19.7b_2

Technique

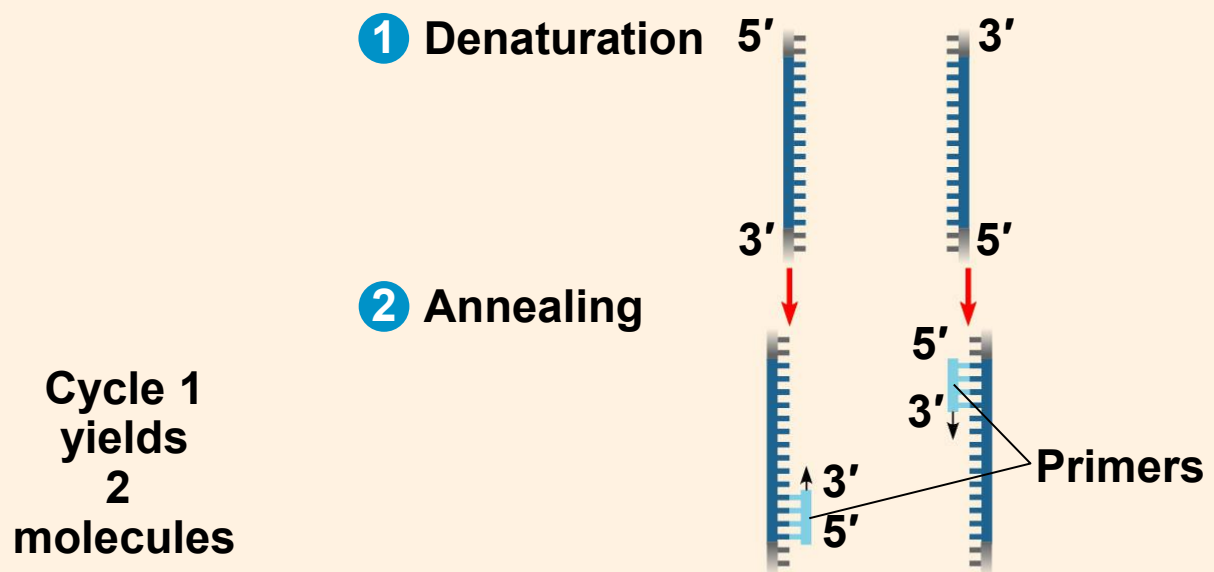


Figure 19.7b_3

Technique

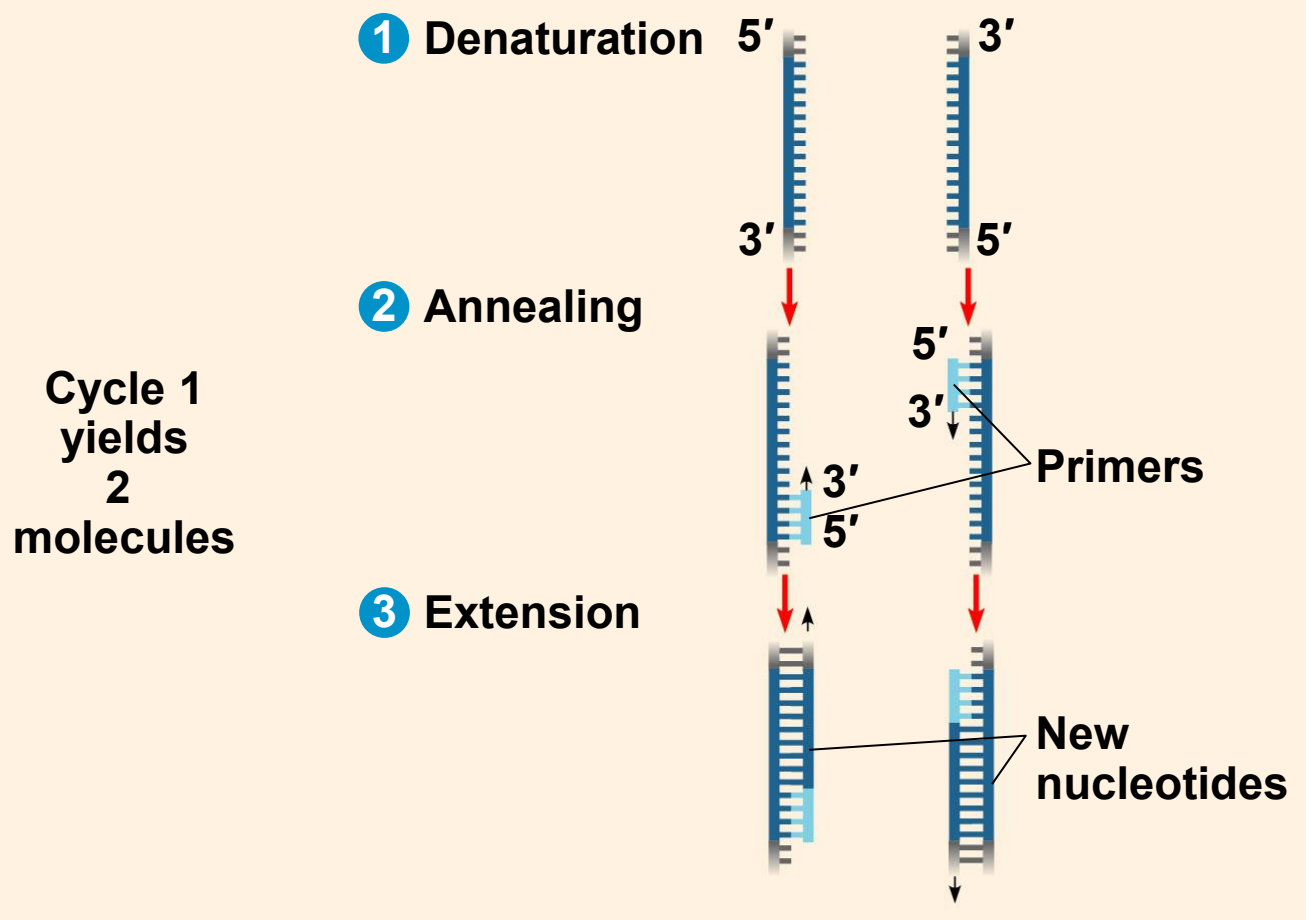
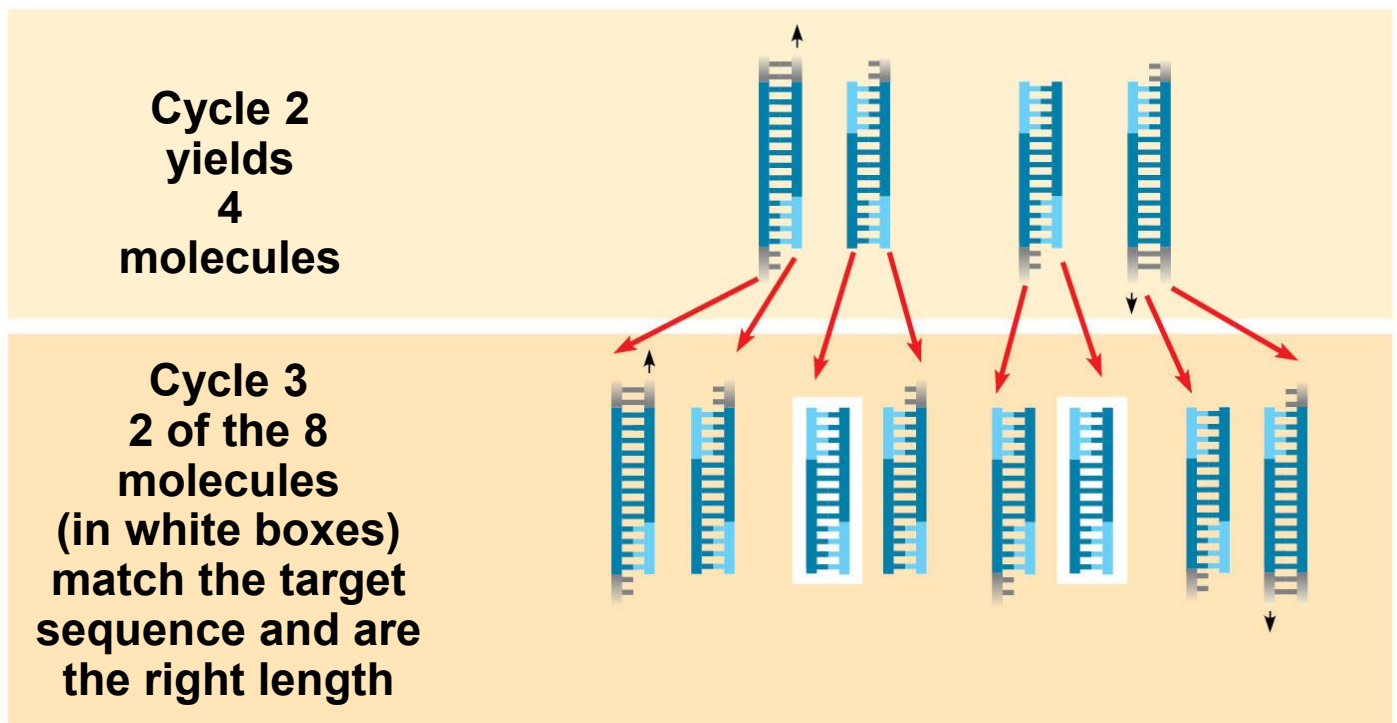


Figure 19.7c

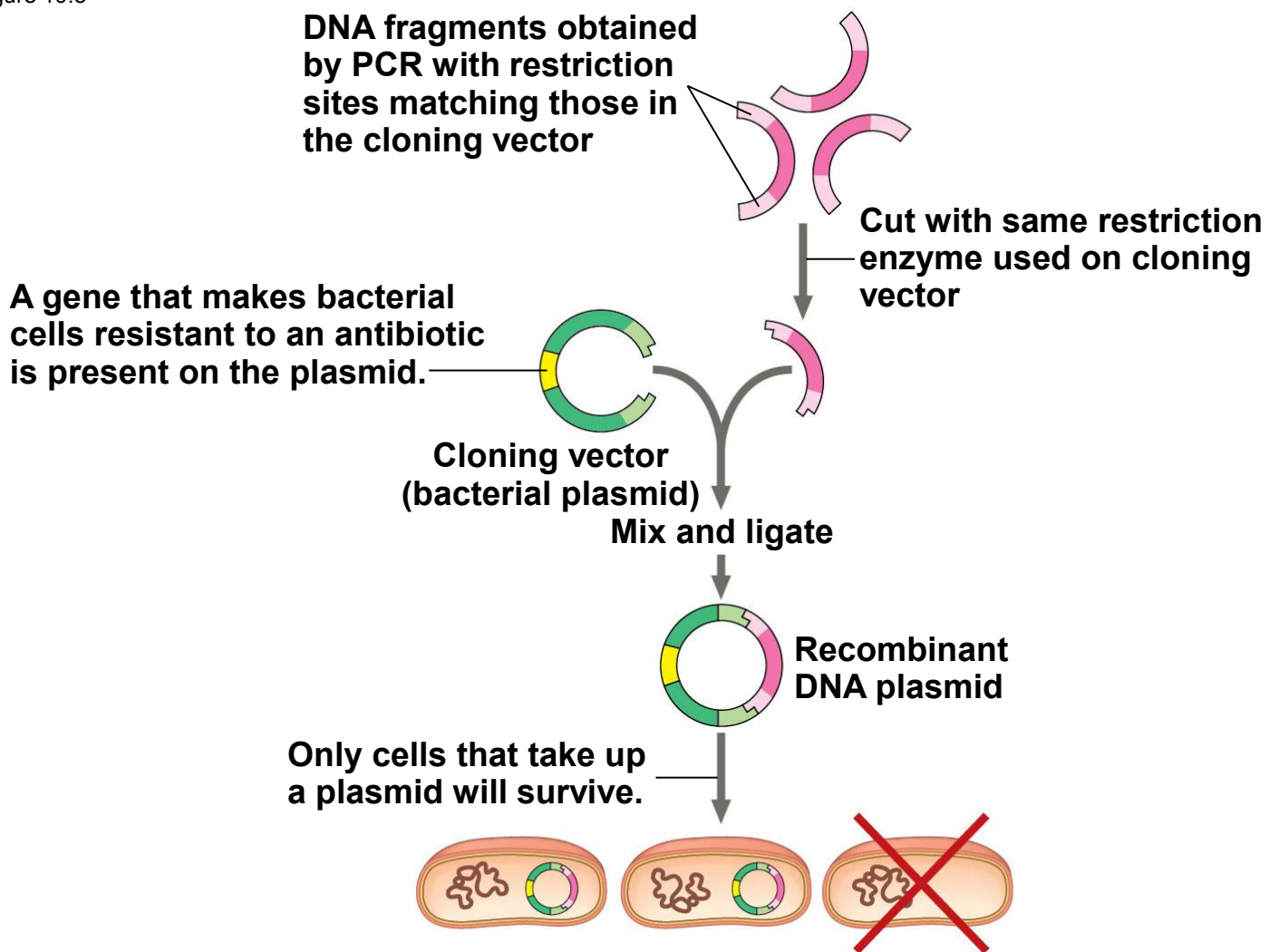
Technique



Results After 30 more cycles, over 1 billion (10^9) molecules match the target sequence.

- PCR primers can be designed to include restriction sites that allow the product to be cloned into plasmid vectors
- The resulting clones are sequenced and error-free inserts selected

Figure 19.8



Expressing Cloned Eukaryotic Genes

- After a gene has been cloned, its protein product can be produced in larger amounts for research or practical applications
- Cloned genes can be expressed in either bacterial or eukaryotic cells

Bacterial Expression Systems

- Several technical difficulties hinder expression of cloned eukaryotic genes in bacterial host cells
- To overcome differences in promoters and other DNA control sequences, scientists usually employ an **expression vector**, a cloning vector that contains a highly active bacterial promoter

- Another problem with eukaryotic gene expression in bacteria is the presence of introns in most eukaryotic genes
- Researchers can avoid this problem by using cDNA, complementary to the mRNA, which contains only exons

Eukaryotic DNA Cloning and Expression Systems

- Molecular biologists can avoid eukaryote-bacterial incompatibility issues by using eukaryotic cells, such as yeasts, as hosts for cloning and expressing genes
- Even yeasts may not be able to modify expressed mammalian proteins properly
- In such cases, cultured mammalian or insect cells may be used to express and study proteins

- One method of introducing recombinant DNA into eukaryotic cells is **electroporation**, applying a brief electrical pulse to create temporary holes in plasma membranes
- Alternatively, scientists can inject DNA into cells using microscopically thin needles
- Once inside the cell, the DNA is incorporated into the cell's DNA by natural genetic recombination

Cross-Species Gene Expression and Evolutionary Ancestry

- The remarkable ability of bacteria to express some eukaryotic proteins underscores the shared evolutionary ancestry of living species
- For example, *Pax-6* is a gene that directs formation of a vertebrate eye; the same gene in flies directs the formation of an insect eye (which is quite different from the vertebrate eye)
- The *Pax-6* genes in flies and vertebrates can substitute for each other

Concept 19.2: Biologists use DNA technology to study gene expression and function

- Analysis of when and where a gene or group of genes is expressed can provide important clues about gene function

Analyzing Gene Expression

- The most straightforward way to discover which genes are expressed in certain cells is to identify the mRNAs being made

Studying the Expression of Single Genes

- mRNA can be detected by nucleic acid hybridization with complementary molecules
- These complementary molecules, of either DNA or RNA, are **nucleic acid probes**

- ***In situ* hybridization** uses fluorescent dyes attached to probes to identify the location of specific mRNAs in place in the intact organism

Figure 19.9

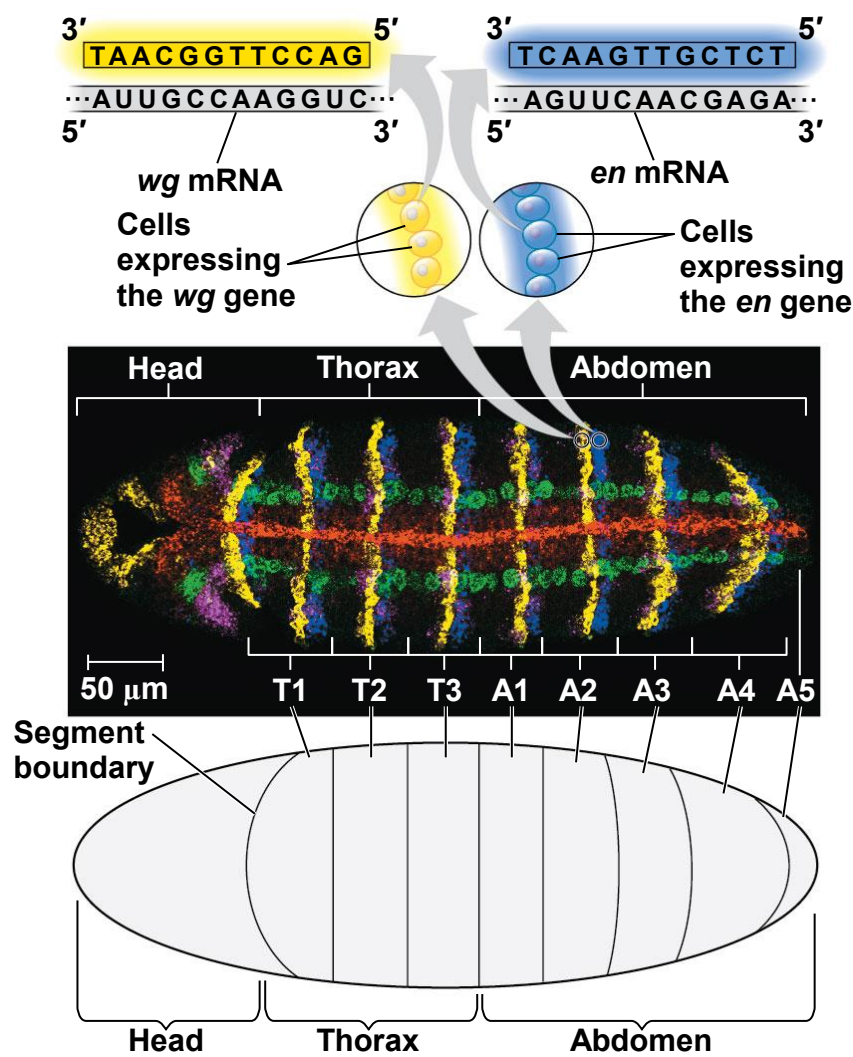


Figure 19.9d

5' ... C U C A U C A C C G G C ... 3'

Figure 19.9e

3' **GAGTAGTGGCCG** 5'

- **Reverse transcriptase-polymerase chain reaction (RT-PCR)** is useful for comparing amounts of specific mRNAs in several samples at the same time
- Reverse transcriptase is added to mRNA to make **complementary DNA (cDNA)**, which serves as a template for PCR amplification of the gene of interest
- The products are run on a gel, and the complement of the mRNA of interest is identified

Figure 19.10_1

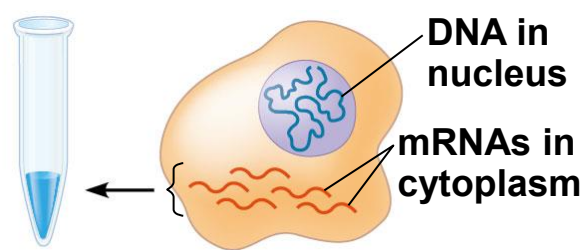


Figure 19.10_2

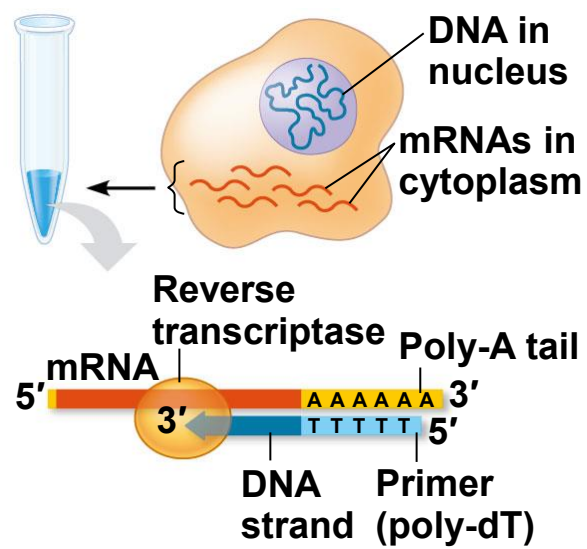


Figure 19.10_3

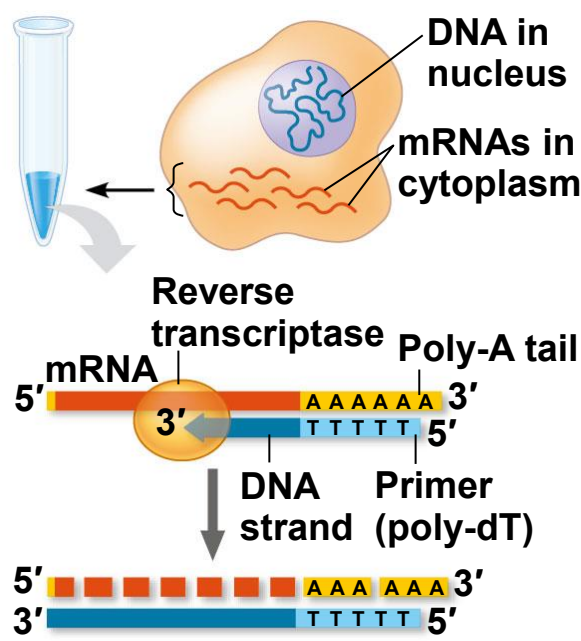


Figure 19.10_4

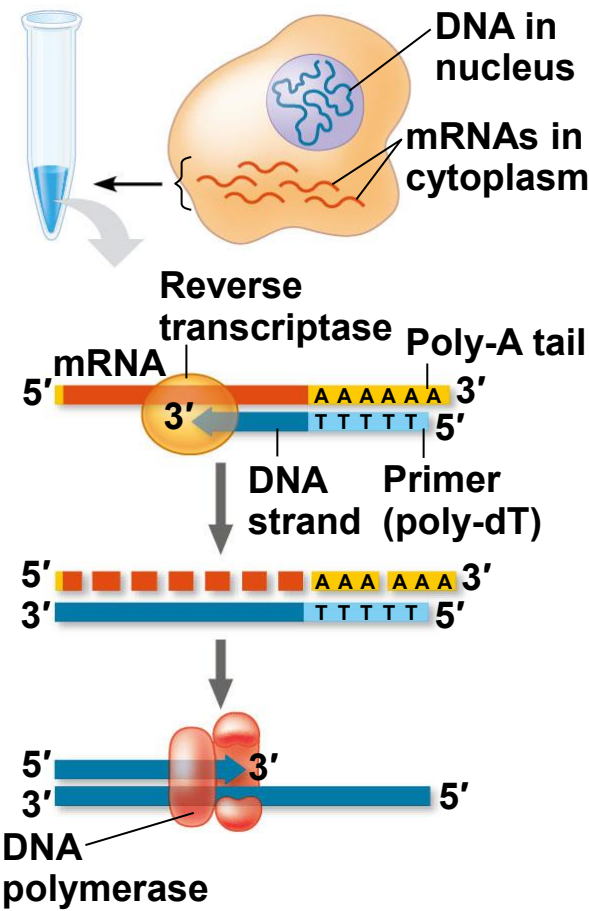
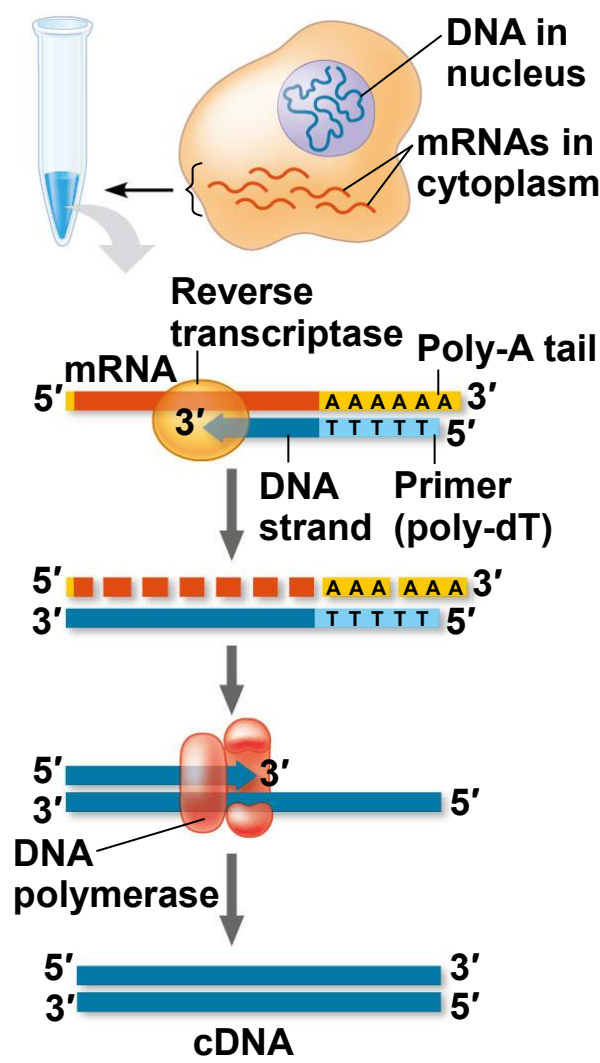


Figure 19.10_5



- An enhancement called quantitative RT-PCR (qRT-PCR) uses a dye that fluoresces only when bound to a double-stranded PCR product
- Newer methods of this type do not require gel electrophoresis
- It can be used to discover which tissue is producing a specific mRNA

Figure 19.11

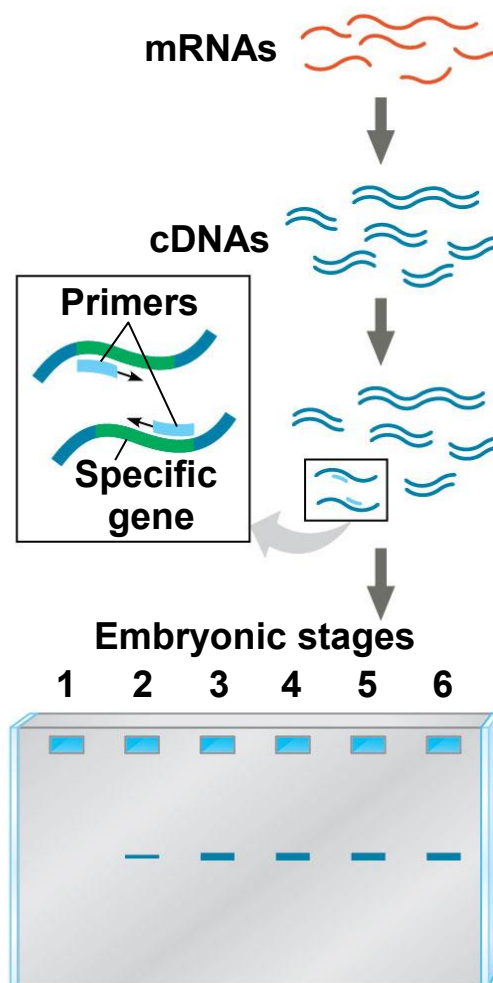
Technique

1 cDNA synthesis

2 PCR amplification

Results

3 Gel electrophoresis



Studying the Expression of Interacting Groups of Genes

- The study of expression of thousands of genes at one time constitutes a systems approach
- One aim is to identify networks of gene expression across an entire genome
- **DNA microarray assays** compare patterns of gene expression in different tissues at different times or under different conditions
- A microarray consists of tiny amounts of many single-stranded genes fixed to a glass slide

Figure 19.12

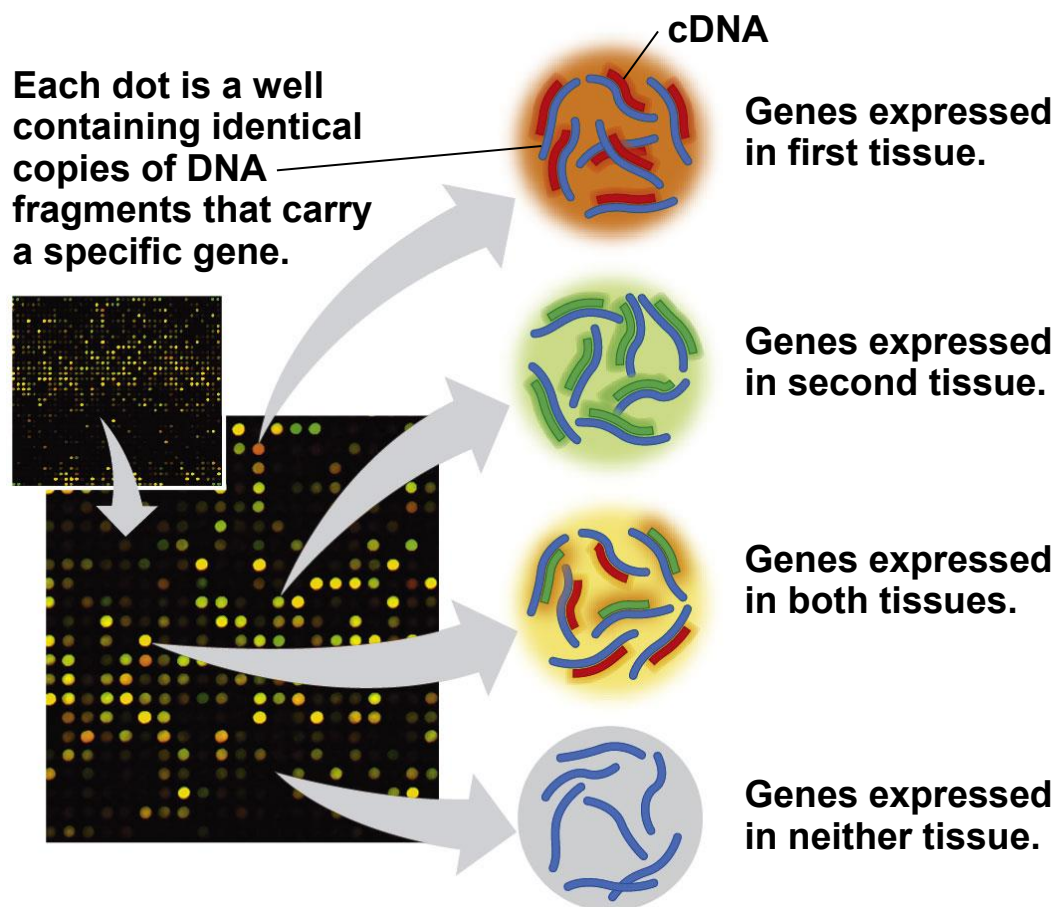
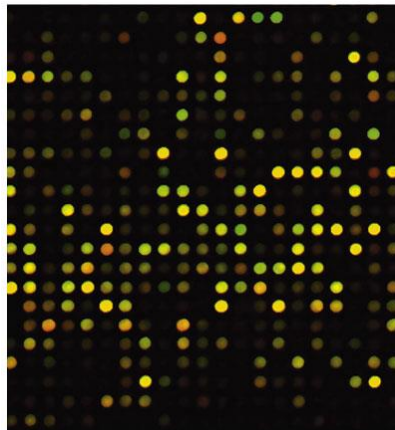


Figure 19.12a

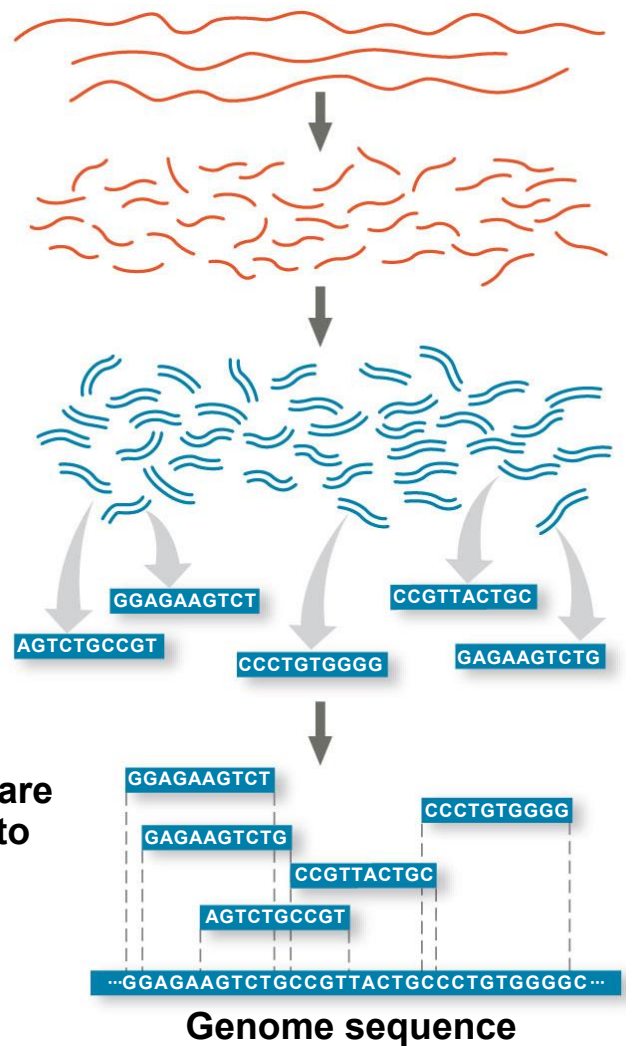
**Each dot is a well
containing identical
copies of DNA
fragments that carry
a specific gene.**



- With rapid and inexpensive sequencing methods, researchers can now just sequence cDNA samples from different tissues or embryonic stages to determine the gene expression differences between them
- This approach is called **RNA sequencing** or **RNA-seq**
- RNAs are isolated, cut into short, similar-sized fragments, converted into cDNAs, and sequenced

Figure 19.13

- 1** mRNAs are isolated from the tissue being studied.
- 2** mRNAs are cut into similar-sized, small fragments.
- 3** mRNAs are reverse-transcribed into cDNAs of the same size.
- 4** cDNAs are sequenced.
- 5** The short sequences are mapped by computer onto the genome sequence.



- Scientists can now measure expression of thousands of genes at one time
- Information from these methods should provide a grander view of how ensembles of genes interact to form an organism and maintain its vital systems

Determining Gene Function

- One way to determine function is to disable the gene and observe the consequences

Editing Genes and Genomes

- Using ***in vitro* mutagenesis**, specific mutations are introduced into a cloned gene, altering or destroying its function
- When the mutated gene is returned to the cell, the normal gene's function might be determined by examining the mutant's phenotype

- The **CRISPR-Cas9 system** is a powerful new technique for gene editing in living cells and organisms
- Cas9 acts together with a “guide RNA” made from the CRISPR region of the bacterial system
- It will cut both strands of any DNA sequence complementary to the guide RNA
- If the guide RNA is engineered to be complementary to the target gene, the target DNA will be cut

- When the cut DNA is repaired, nucleotides may be introduced or removed, causing the gene to be inactivated
- Researchers have also modified the technique so a gene with a mutation in it can be repaired
- This is done by introducing a segment of the wild-type gene, which may be used as a template to repair the target DNA

Figure 19.14

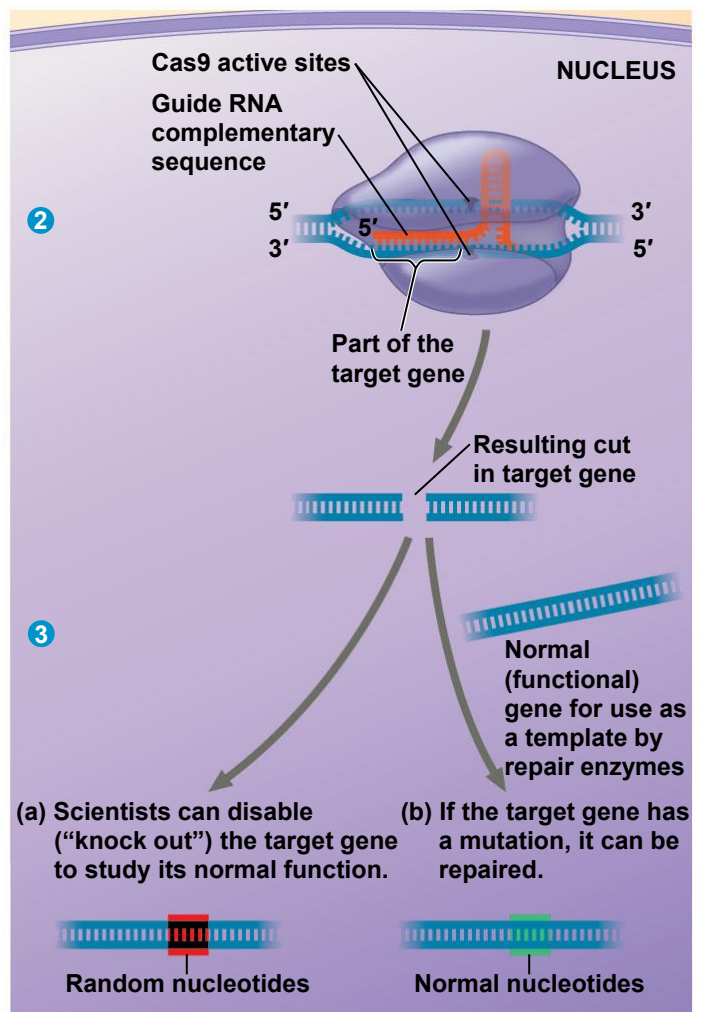
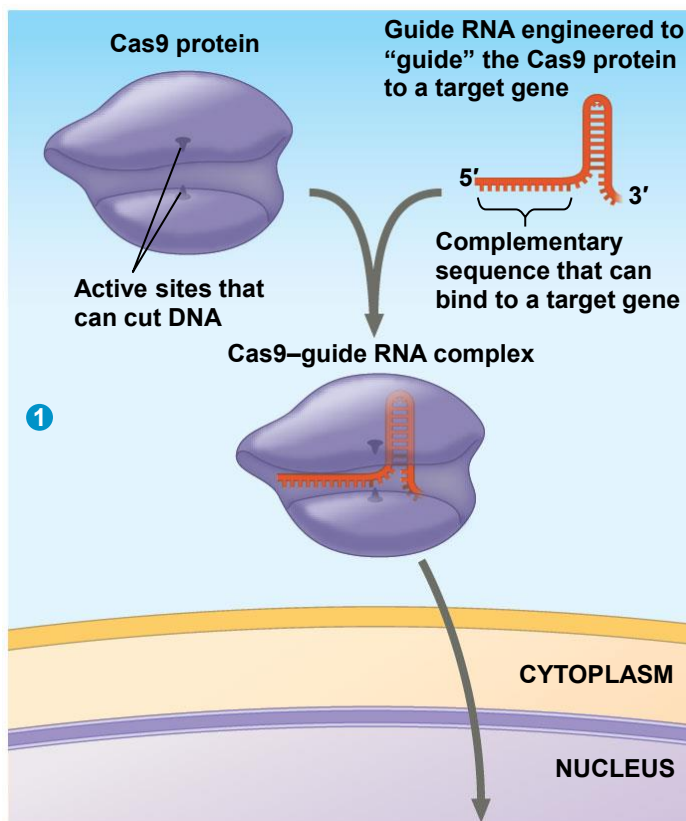


Figure 19.14a

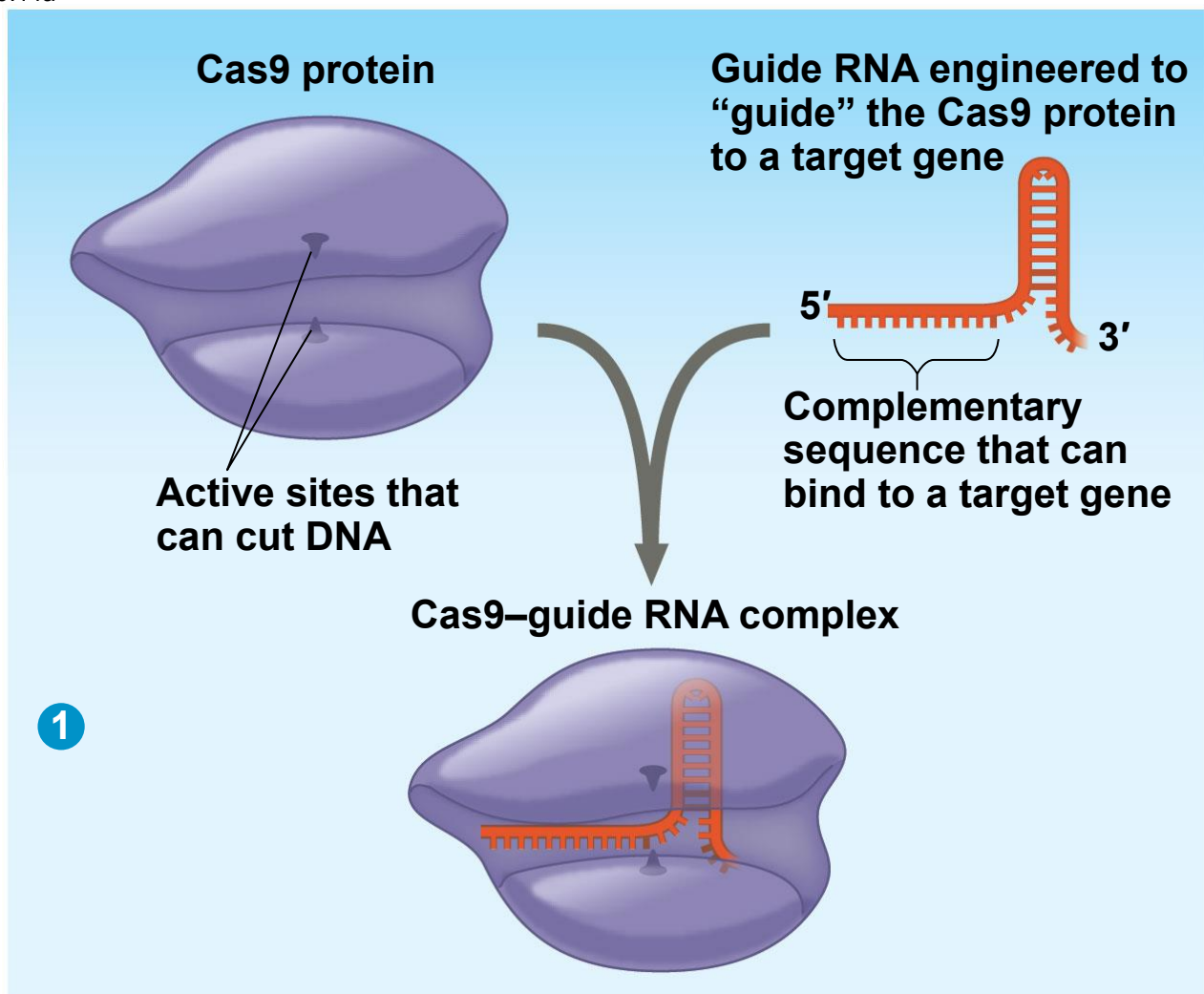


Figure 19.14b

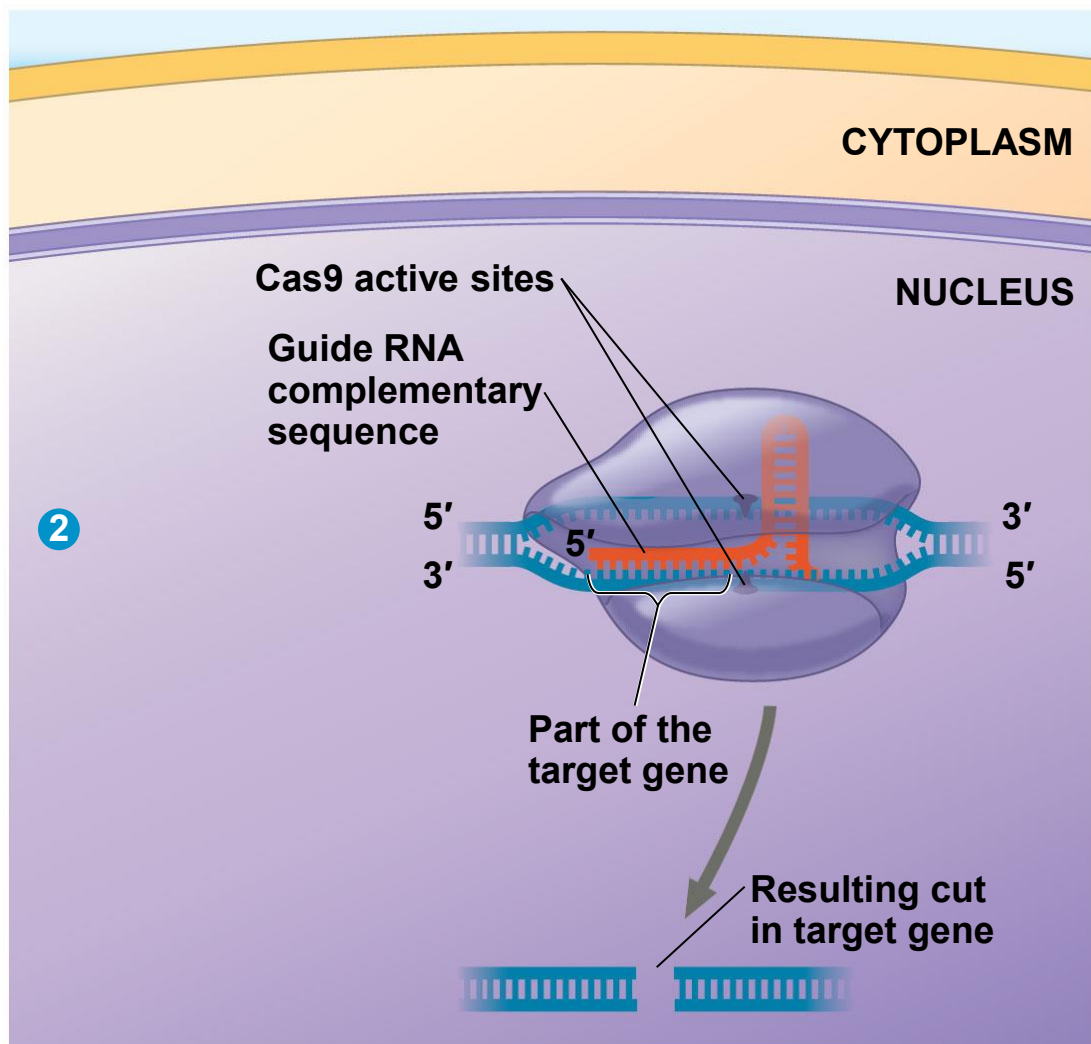
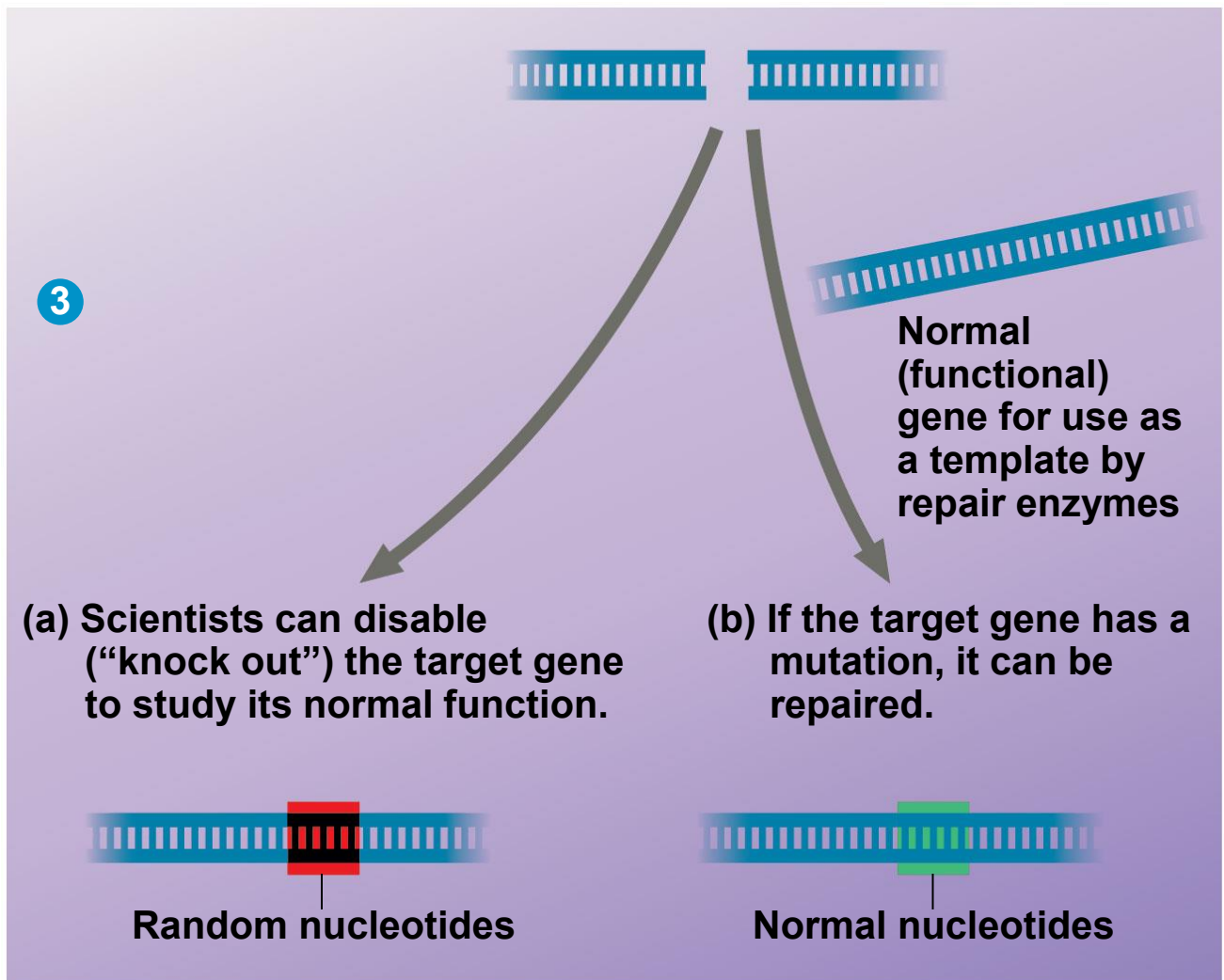


Figure 19.14c



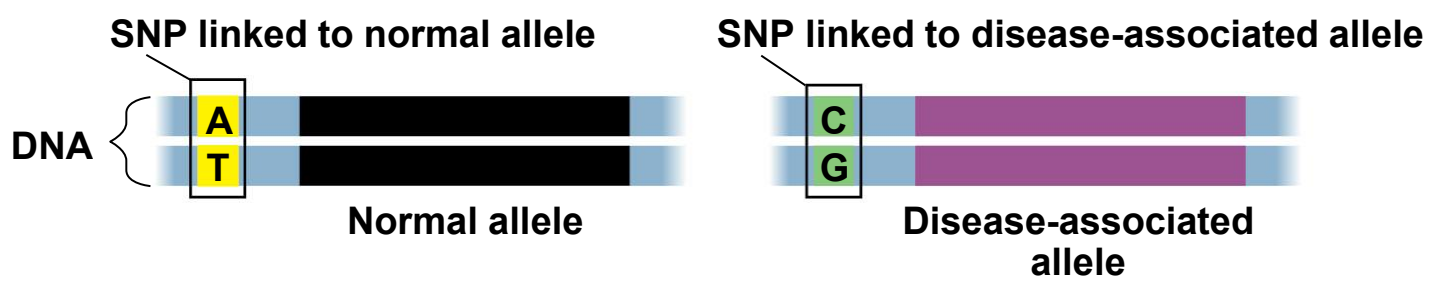
Other Methods for Studying Gene Function

- Gene expression can also be silenced using **RNA interference (RNAi)**
- Synthetic double-stranded RNA molecules matching the sequence of a particular gene are used to break down or block the gene's mRNA

- In humans, researchers analyze the genomes of many people with a certain genetic condition to try to find nucleotide changes specific to the condition
- These **genome-wide association studies** test for genetic markers, sequences that vary among individuals
- **SNPs (single nucleotide polymorphisms)**, single nucleotide variants, are among the most useful genetic markers

- SNPs that are found frequently associated with a particular inherited disorder alert researchers to the most likely location for the disease-causing gene
- SNPs are rarely directly involved in the disease; they are most often in noncoding regions of the genome

Figure 19.15



Concept 19.3: Cloned organisms and stem cells are useful for basic research and other applications

- Organismal cloning produces one or more organisms genetically identical to the “parent” that donated the single cell
- A **stem cell** is a relatively unspecialized cell that can reproduce itself indefinitely, or under certain conditions can differentiate into one or more types of specialized cells

Cloning Plants: Single-Cell Cultures

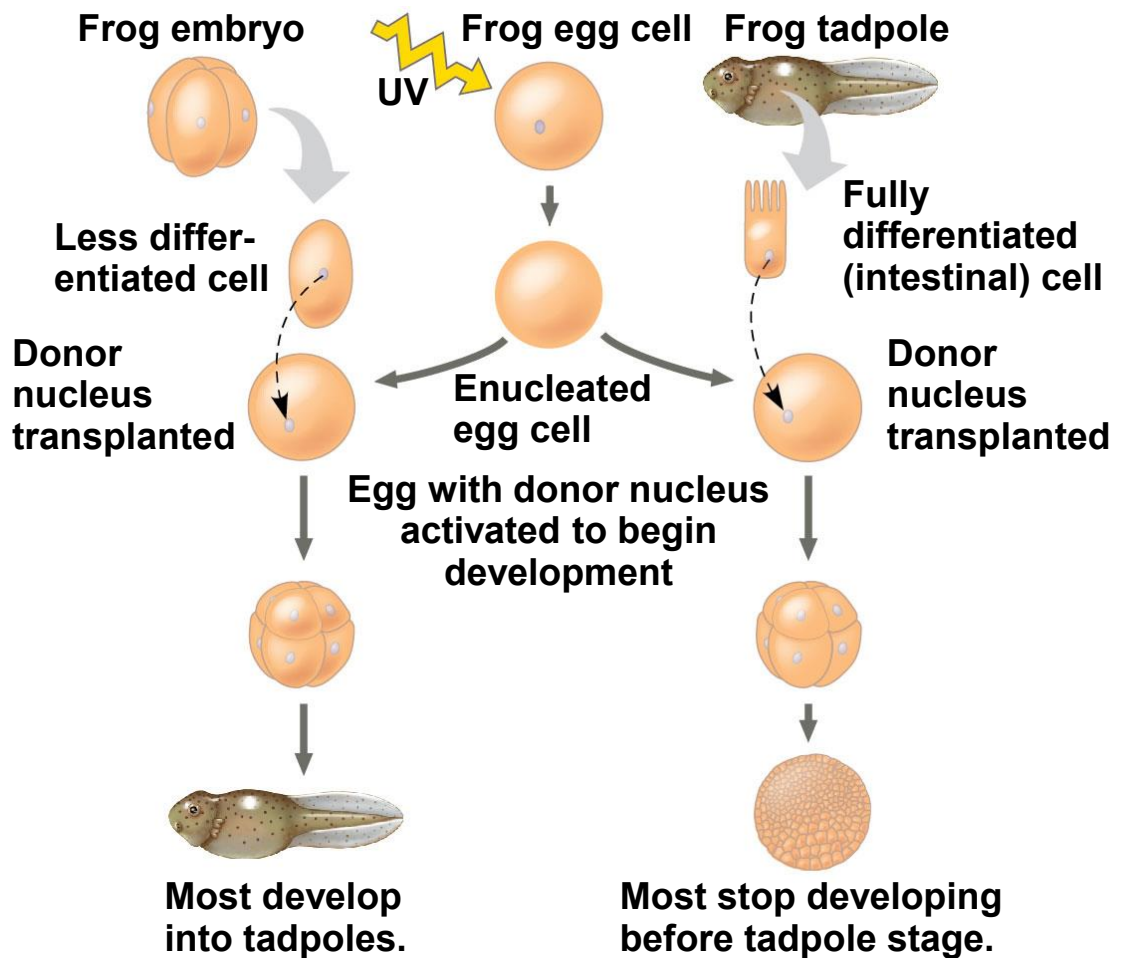
- In plants, mature cells can dedifferentiate and then give rise to all the specialized cell types of the organism
- A **totipotent** cell, such as this, is one that can generate a complete new organism
- Plant cloning is used extensively in agriculture

Cloning Animals: Nuclear Transplantation

- In nuclear transplantation, the nucleus of an unfertilized egg cell or zygote is replaced with the nucleus of a differentiated cell
- Experiments with frog embryos have shown that a transplanted nucleus can often support normal development of the egg
- However, the older the donor nucleus, the lower the percentage of normally developing tadpoles

Figure 19.16

Experiment



Results

Data from J. B. Gurdon et al., The developmental capacity of nuclei transplanted from keratinized cells of adult frogs, *Journal of Embryology and Experimental Morphology* 34:93–112 (1975).

Reproductive Cloning of Mammals

- In 1997, Scottish researchers announced the birth of Dolly, a lamb cloned from an adult sheep by nuclear transplantation from a differentiated mammary cell
- Dolly's premature death in 2003, as well as that of another cloned sheep from another experiment, led to speculation that her cells were not as healthy as those of a normal sheep
- This possibly reflects incomplete reprogramming of the original transplanted nucleus

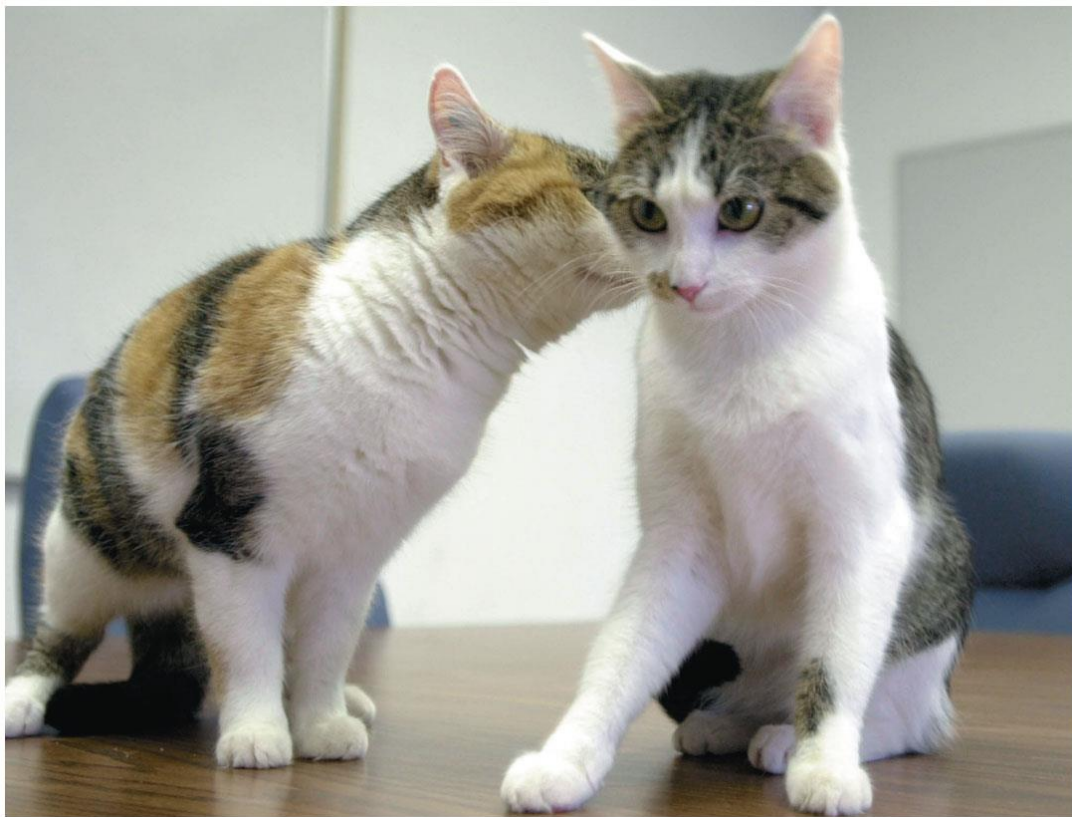
Figure 19.17



© Photo courtesy of the Roslin Institute,
The University of Edinburgh

- Since 1997, cloning has been demonstrated in many mammals, including mice, cats, cows, horses, mules, pigs, and dogs
- CC (for Carbon Copy) was the first cat cloned; however, CC differed somewhat from her female “parent”
- Cloned animals do not always look or behave exactly the same
- Even identical human twins, natural “clones,” are always slightly different

Figure 19.18



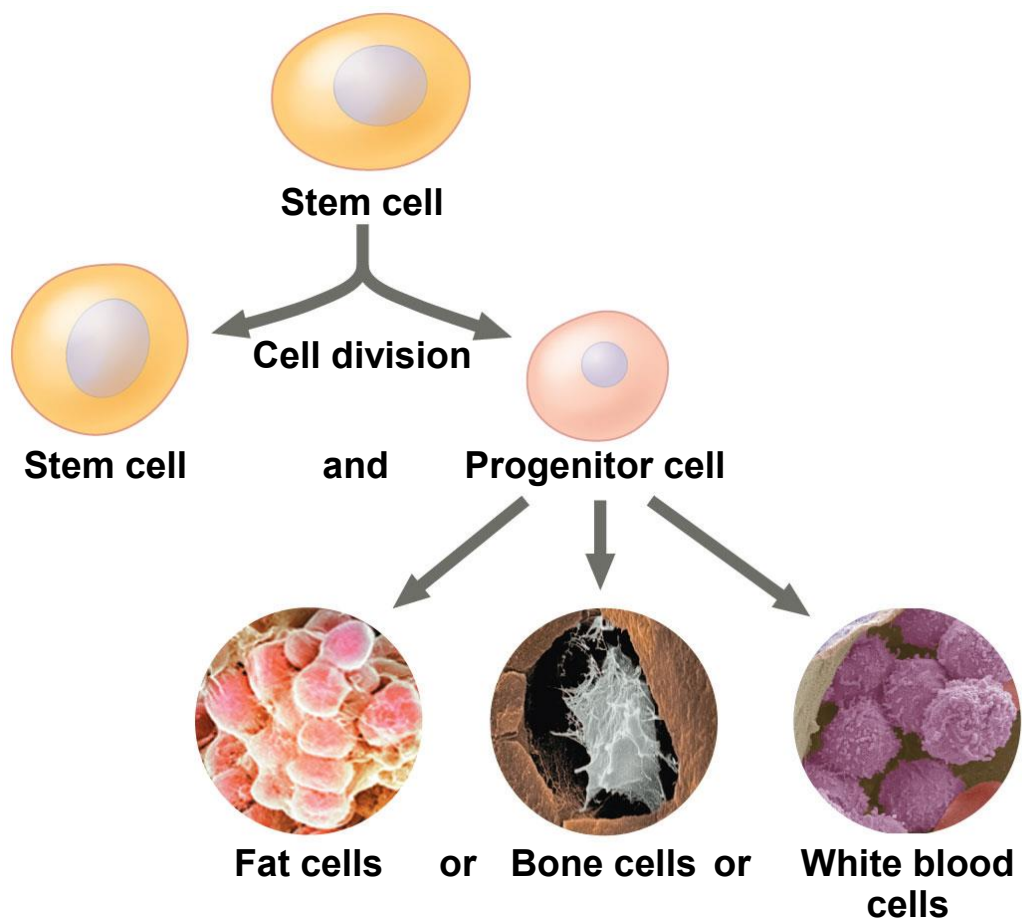
Faulty Gene Regulation in Cloned Animals Due to Epigenetic Differences

- In most nuclear transplantation studies, only a small percentage of cloned embryos have developed normally to birth
- Many cloned animals exhibit defects
- Many epigenetic changes, such as acetylation of histones or methylation of DNA, must be reversed in the nucleus from a donor animal in order for genes to be expressed or repressed appropriately for early stages of development

Stem Cells of Animals

- Stem cells are relatively unspecialized cells that can both reproduce indefinitely and, under certain conditions, differentiate into one or more specialized cell types

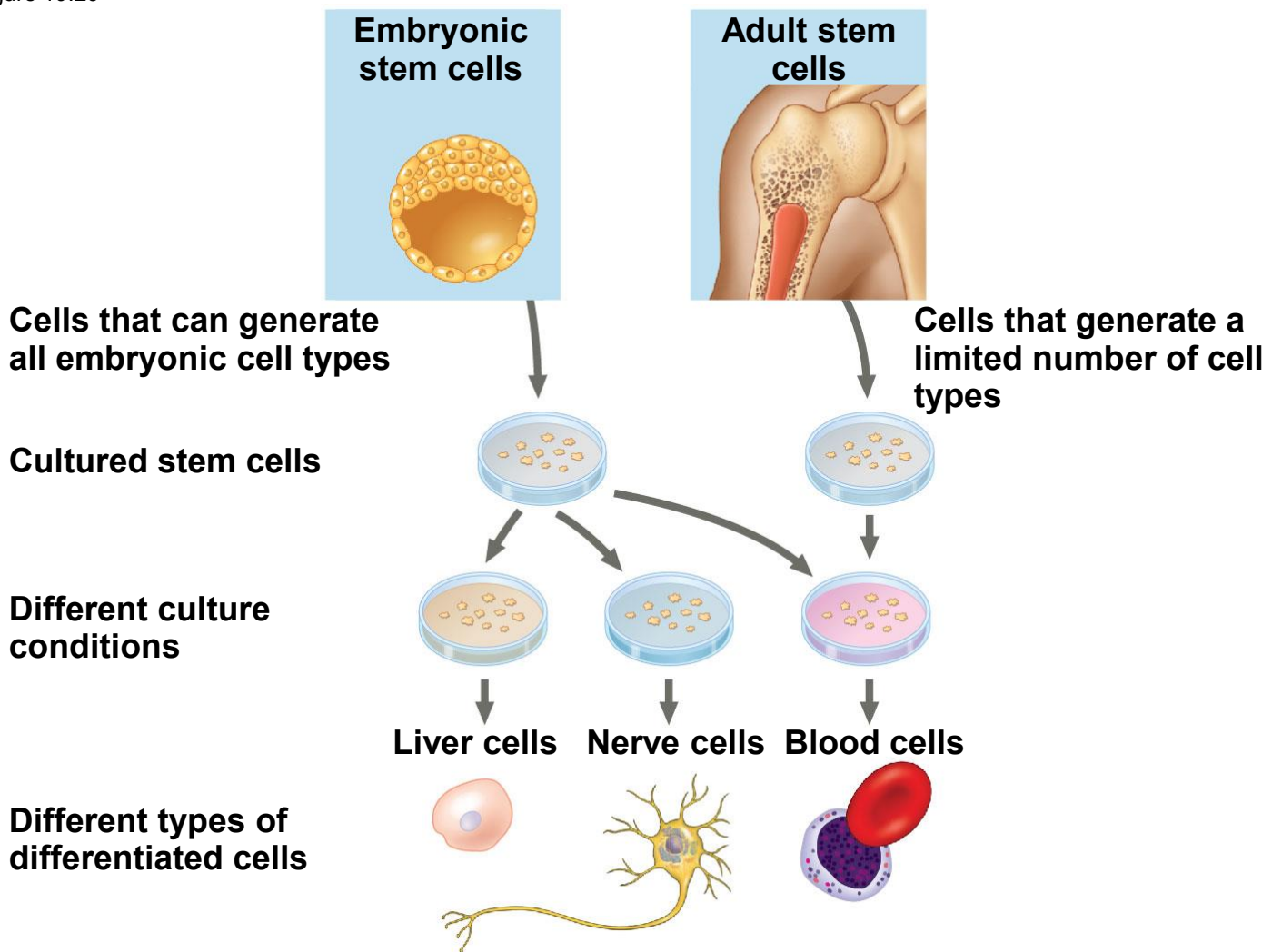
Figure 19.19



Embryonic and Adult Stem Cells

- Many early embryos contain stem cells capable of giving rise to differentiated embryonic cells of any type
- In culture, these embryonic stem (ES) cells reproduce indefinitely, and depending on culture conditions, can be made to differentiate into a variety of specialized cells
- Adult stem cells can generate multiple (but not all) cell types and are used in the body to replace nonreproducing cells as needed

Figure 19.20



- Embryonic stem cells are **pluripotent**, capable of differentiating into many different cell types
- Originally, ES cells were obtained only from embryos donated by patients undergoing infertility treatments
- The techniques for cloning human embryos are still being optimized, but they represent a potential new (less controversial) source of ES cells

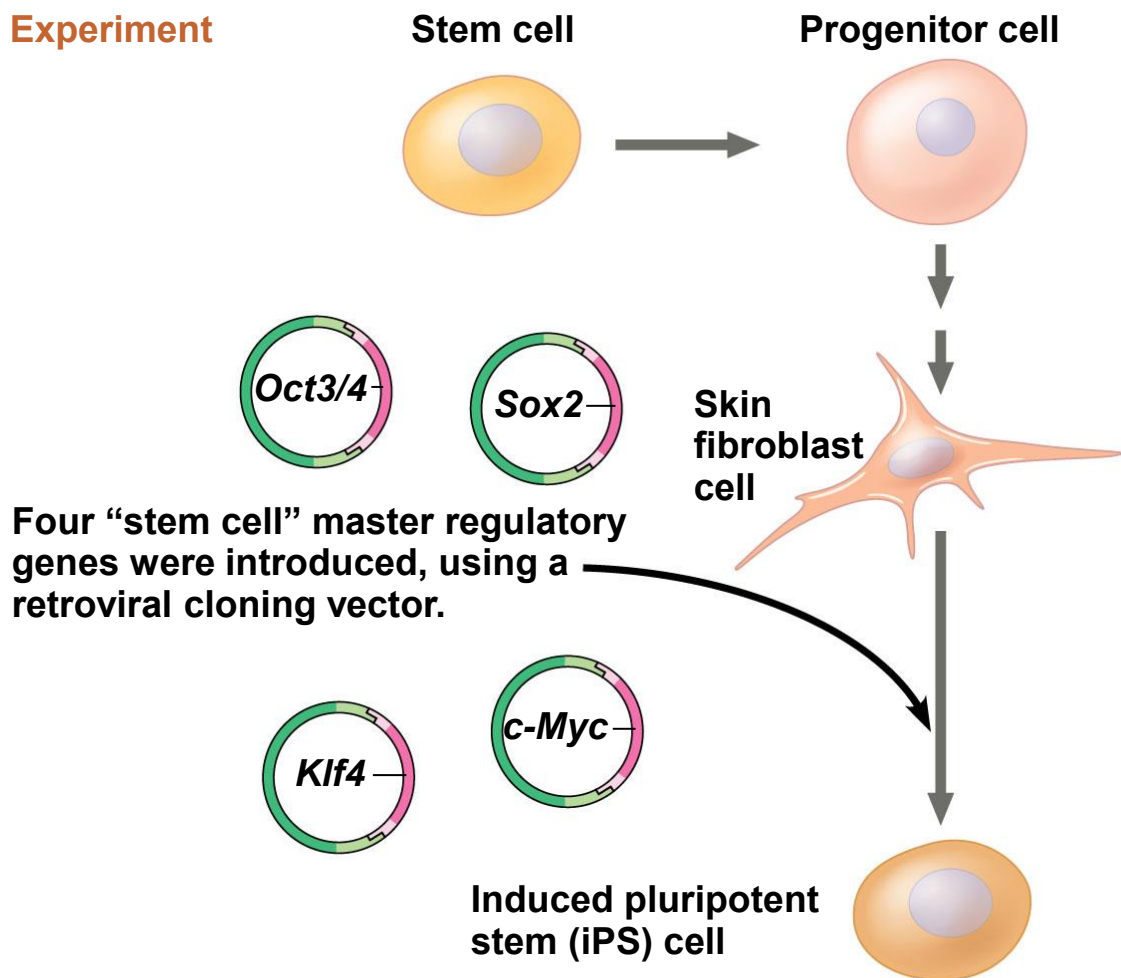
- The main aim of cloning ES cells is to produce cells for treating disease
- The process is thus called therapeutic cloning
- Opinions vary about the morality of therapeutic cloning

Induced Pluripotent Stem (iPS) Cells

- Researchers can treat differentiated cells and reprogram them to act like ES cells
- Researchers used retroviruses to induce extra copies of four stem cell master regulatory genes to produce induced pluripotent stem (iPS) cells
- iPS cells can perform most of the functions of ES cells
- iPS cells can be used as models for study of certain diseases and potentially as replacement cells for patients

Figure 19.21

Experiment



Data from K. Takahashi et al., Induction of pluripotent stem cells from adult human fibroblasts by defined factors, *Cell* 131:861–872 (2007).

- Human iPS lines have been developed from individuals with type 1 diabetes, Parkinson's disease, Huntington's disease, Down syndrome, and other diseases
- Researchers have also identified genes that can reprogram a differentiated cell into another type of differentiated cell without it passing through a pluripotent state
- Cells created this way could be used without raising most ethical objections

- As biotechnology continues to change, so does its use in agriculture, industry, and medicine
- Great benefits could result from biotechnology approaches, but unforeseen problems could arise
- We must proceed with humility and caution